

72nd BPSC PRELIMS

प्रहार

Test Series

Test-17/1

FULL LENGTH TEST

Solution

English



BPSC TEST-17 (SOLUTION)

Q 1. Ans. D

Explanation:

- In a world-first announced in 2025, the world's first cryo-born baby corals were settled on the Great Barrier Reef in Australia.
- Reproductive biologists from Taronga Conservation Society Australia, working with the Australian Institute of Marine Science, fertilised fresh coral eggs (collected during the October spawning) using cryopreserved coral sperm from the Taronga CryoDiversity Bank, the world's largest frozen repository of living coral cells.
- The resulting larvae were grown at the National Sea Simulator and then deployed in specially designed "coral cradles" on the Reef.
- The aim is to enable large-scale planting of heat-tolerant corals to help reefs withstand climate change and bleaching.

Q 2. Ans. A

Explanation:

- The "Global Cybersecurity Outlook" is a flagship annual publication of the World Economic Forum (WEF), produced in collaboration with Accenture.
- It is released every January ahead of the annual Davos meeting and examines the cybersecurity trends, threats, and priorities expected to affect economies and societies in the year ahead.
- The most recent edition, Global Cybersecurity Outlook 2026 (now in its fifth year), focused on AI acceleration, geopolitical fragmentation, and widening cyber inequity.

Q 3. Ans. B

Explanation:

- Norovirus is the world's most common cause of acute gastroenteritis and is distinguished from most bacterial foodborne pathogens (such as Salmonella, Campylobacter, and Clostridium perfringens) precisely by its remarkably efficient person-to-person transmissibility.
- Ingesting as few as 18 viral particles is enough to cause infection.
- Transmission occurs through the faecal-oral route, by contact with contaminated surfaces, through contaminated food or water, and even via aerosolised vomit particles.

- This is why Norovirus outbreaks spread so rapidly in closed settings such as cruise ships, hospitals, schools, and care homes in ways that most bacterial foodborne infections do not.

Q 4. Ans. C

Explanation:

- **Statement 1 is correct:** IOMed is formally described as the world's first intergovernmental legal organisation dedicated exclusively to the resolution of international disputes through mediation. This was confirmed in the official Convention text and repeatedly stated by the Chinese Foreign Ministry, the Hong Kong SAR government, and the IOMed's own website at the time of the signing ceremony.
- **Statement 2 is incorrect:** The Convention on the Establishment of IOMed was signed on 30 May 2025 in Hong Kong, not Taipei. The signing ceremony was held at the Grand Hyatt Hong Kong, with 33 founding signatory states participating. IOMed is also headquartered in Hong Kong.
- **Statement 3 is incorrect:** The Convention does not require ratification by all signatory states to enter into force. It enters into force following ratification, acceptance, or approval by at least three signatories, making the threshold far lower than universal ratification.
- **Statement 4 is correct:** IOMed is designed to offer mediation as an additional and distinct pathway alongside existing international mechanisms. It explicitly positions itself as implementing Article 33 of the UN Charter, which lists mediation as one among several peaceful means of settling disputes (alongside arbitration, judicial settlement, and others). Rather than displacing bodies such as the ICJ or ICSID arbitration, it provides a complementary party-driven, consent-based alternative.

Q 5. Ans. C

Explanation:

- Scurvy is caused by a prolonged dietary deficiency of Vitamin C (ascorbic acid).
- Vitamin C is essential for the synthesis of collagen; its deficiency causes bleeding and swollen gums, joint pain, and delayed wound healing.
- Rich sources of Vitamin C include citrus fruits such as oranges and lemons, amla, and guava.
- Vitamin A deficiency causes night blindness and

xerophthalmia, Vitamin B1 deficiency causes beriberi, and Vitamin D deficiency causes rickets in children.

Q 6. Ans. B

Explanation:

- **Monera (a - 2):** The kingdom Monera consists entirely of **prokaryotic organisms**. These are the most primitive life forms; they lack a true, membrane-bound nucleus and other cellular organelles. (Examples: Bacteria, Cyanobacteria/Blue-green algae, and Mycoplasma).
- **Protista (b - 1):** The kingdom Protista includes primarily **unicellular eukaryotes**. Unlike Monerans, Protists have a well-defined nucleus and membrane-bound organelles. They often possess appendages like cilia or flagella for movement. (Examples: Amoeba, Paramecium, Euglena, and Diatoms).
- **Binomial nomenclature (c - 3):** This is the formal system of naming species using a **two-part scientific name**. The first part identifies the Genus (capitalized) and the second part identifies the species (lowercase). It was introduced by **Carolus Linnaeus**, who is known as the father of modern taxonomy. (Example: Homo sapiens).
- **Robert Whittaker (d - 4):** R.H. Whittaker was the American ecologist who proposed the widely accepted **Five-Kingdom Classification** in 1969. He categorized all living organisms into Monera, Protista, Fungi, Plantae, and Animalia based on cell structure, body organization, mode of nutrition, and phylogenetic relationships.

Q 7. Ans. C

Explanation:

- **Assertion (A) is correct:** **Auxin** is a primary plant growth hormone synthesized at the growing tips of stems and roots. Its main function is to promote cell elongation, which helps the plant shoot grow longer. It is also the hormone responsible for phototropism (plants bending towards the light)—when light hits one side of a shoot, auxin accumulates on the shaded side, causing those cells to elongate faster and bend the plant toward the light source.
- **Reason (R) is incorrect:** **Abscisic acid (ABA)** is **not** a growth-promoting hormone; it is the exact opposite. It is a fundamental growth inhibitor. Often called the “stress hormone,” ABA halts growth during unfavorable environmental conditions, promotes seed and bud dormancy, and triggers the wilting and falling of leaves. It also causes the stomata to close during severe droughts to prevent water loss.

Q 8. Ans. A

Explanation:

- **Statement 1 is correct:** The **nephron** is indeed the fundamental structural and functional filtration unit of the human kidney. Each kidney contains over a million of these microscopic filtering units, which are responsible for cleaning the blood by removing nitrogenous wastes (like urea) and excess water to form urine.
- **Statement 2 is correct:** **Guard cells** are specialized, bean-shaped epidermal cells that flank the stomatal pores on the surface of plant leaves. They regulate the opening and closing of stomata by swelling (when water flows into them, opening the pore for gas exchange and transpiration) and shrinking (when water is lost, closing the pore to prevent dehydration).
- **Statement 3 is incorrect:** Bile is **not** secreted by the pancreas. Bile juice is secreted by the **Liver** (the largest gland in the body) and is temporarily stored in the gallbladder. While bile is essential for the digestion of fats (a process called emulsification, which breaks down large fat globules into smaller ones), the pancreas secretes pancreatic juice, which contains different digestive enzymes like trypsin (for proteins) and lipase (for breaking down the emulsified fats).

Q 9. Ans. B

Explanation:

- **Pair 1 is correctly matched:** Genotype refers to the exact genetic makeup or constitution of an organism. It is the specific set of alleles an individual carries (e.g., TT, Tt, or tt for plant height).
- **Pair 2 is correctly matched:** Phenotype refers to the observable physical characteristics or traits of an organism, which are determined by the interaction between its genotype and the environment (e.g., whether the plant is visibly tall or short).
- **Pair 3 is incorrectly matched:** Homologous organs have the same basic anatomical structure and embryonic origin, but they perform different functions. They indicate divergent evolution.
 - ✓ **Example:** The forelimbs of humans (grasping), whales (swimming), and bats (flying) share the same underlying bone structure (humerus, radius, ulna) but do completely different jobs.
- **Pair 4 is incorrectly matched:** Analogous organs have different basic anatomical structures and origins, but they perform similar functions. They indicate convergent evolution.

- ✓ **Example:** The wings of a bird (made of bones and feathers) and the wings of a butterfly (made of chitinous membranes) look very different structurally, but both are used for the same function, flying.

Q 10. Ans. A

Explanation:

- **Statement 1 is correct:** A hinge joint allows movement primarily in only one plane (uniaxial movement), operating much like the metal hinge of a door. It strictly permits flexion and extension (bending and straightening). Classic examples in the human body include the elbow joint, the knee joint, and the interphalangeal joints (the joints in your fingers and toes).
- **Statement 2 is incorrect:** The joints between the flat bones of the human skull (cranial bones) are fixed joints (specifically called fibrous joints or sutures). They are tightly interlocked and do not allow any movement, serving the crucial function of protecting the brain. A ball and socket joint, on the other hand, is the most mobile joint in the body, allowing multiaxial movement in almost all directions. It is found in the shoulder (where the head of the humerus fits into the glenoid cavity) and the hip (where the femur fits into the acetabulum of the pelvis).
- **Pivot Joint:** Allows for rotational movement around a single axis. The classic example is the joint between the first two cervical vertebrae (the atlas and the axis), which allows you to turn your head side-to-side to say “no.”
- **Gliding (Planar) Joint:** Allows flat or nearly flat bones to slide or glide past one another. These are found between the carpal bones in the wrist and the tarsal bones in the ankle.

Q 11. Ans. D

Explanation:

- Human teeth are heterodont (meaning we have different types of teeth for different functions), and **canines** are the sharp, pointed teeth situated just right outside the incisors, specifically adapted for tearing and piercing food (like tough meats).
- **Incisors:** These are the flat, chisel-shaped front teeth. They are adapted primarily for **biting and cutting** food into manageable pieces.
- **Premolars (Bicuspid):** Located behind the canines, these have a flatter surface with ridges and are used for **crushing and grinding** food.

- **Molars:** These are the largest teeth at the back of the mouth, featuring a wide, flat surface designed for intense **chewing and fine grinding** before swallowing.

Q 12. Ans. C

Explanation:

- **Assertion (A) is correct:** Amphibians (like frogs and salamanders) and reptiles (like snakes, lizards, and turtles) are indeed **poikilothermic** or “cold-blooded” animals.
- **Reason (R) is incorrect:** Poikilothermic animals do **not** maintain a constant internal body temperature. Instead, their body temperature fluctuates dynamically with the temperature of their surrounding environment. Because they cannot generate enough internal heat, they rely on behavioral adaptations to regulate their temperature, such as basking in the sun to warm up or hiding in the shade to cool down.
- **Homeotherms (Warm-blooded):** Animals that do maintain a constant internal body temperature regardless of the external environment are called homeotherms (or endotherms). Only **birds (Aves) and mammals** fall into this category.
- **Heart Structure Connection:** A major physiological reason amphibians and reptiles are cold-blooded is their heart structure. Amphibians have 3-chambered hearts, and most reptiles have 3-chambered hearts (except crocodiles, which have 4). This results in a mixing of oxygenated and deoxygenated blood, making their metabolic heat production too inefficient to maintain a constant body temperature.

Q 13. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Earth is not a perfect geometric sphere; it is an oblate spheroid, meaning it bulges at the equator and is flattened at the poles. Because the equatorial radius is larger than the polar radius (by about 21 km), and the acceleration due to gravity (g) is inversely proportional to the square of the distance from the center ($g = GM / R^2$), the value of g is minimum at the equator and maximum at the poles. Furthermore, the centrifugal force generated by Earth’s rotation also counters gravity most strongly at the equator, further reducing g there.
- **Statement 2 is correct:** The weight of an object is defined as the gravitational force acting upon

its mass ($W = mg$). Because an object's mass (m) remains constant regardless of location, its weight is directly proportional to the local acceleration due to gravity (g). Since g reaches its maximum value at the poles, an object will correspondingly weigh the maximum at the poles.

● **Other facts:**

- ✓ g decreases as you go high above the Earth's surface.
- ✓ g decreases as you go deep inside the Earth (becoming zero at the absolute center).
- ✓ g increases as you move from the equator toward the poles.

Q 14. Ans. B

Explanation:

- **Statement 1 is correct:** In 2025, Baby KJ Muldoon became the first patient to receive a rapid, hyper-personalized in-vivo CRISPR gene-editing therapy. He was born with severe carbamoyl phosphate synthetase 1 (CPS1) deficiency, an ultra-rare and often fatal metabolic urea cycle disorder.
- **Statement 2 is correct:** The term in-vivo means the therapy was delivered directly into the patient's body to perform the genetic edits. Rather than extracting cells, editing them in a lab, and re-infusing them (which is known as ex-vivo editing), doctors injected lipid nanoparticles carrying the CRISPR base editor directly into his bloodstream to target and edit his liver cells.
- **Statement 3 is incorrect:** CRISPR technology had already been approved by regulatory bodies like the FDA well before KJ's treatment (most notably, the landmark approval of Casgevy for Sickle Cell Disease in 2023). KJ's case was historic not because it was the first CRISPR approval overall, but because it was the first time an on-demand, bespoke CRISPR therapy was rapidly developed and approved for a single, individual patient.
- **Statement 4 is correct:** Because his condition was caused by an ultra-rare, unique single-base mutation (Q335X), standard therapies would not work. The research team sequenced his genome and explicitly engineered the CRISPR guide RNA and base editor to correct his specific genetic "spelling error" within just six months.

Q 15. Ans. A

Explanation:

- DRDO's long-duration scramjet combustor test is a critical technological milestone directly associated

with the development of **Hypersonic Cruise Missiles (HCMs)**.

- A Ramjet engine operates efficiently at supersonic speeds (up to Mach 3 or 4) but chokes at hypersonic speeds. A Scramjet (Supersonic Combustion Ramjet) is an advanced engine designed to maintain supersonic airflow right through the combustion chamber. This allows the vehicle to sustain flight at incredibly high speeds, Mach 5 (five times the speed of sound) and beyond.
- India has been actively developing this capability under the Hypersonic Technology Demonstrator Vehicle (HSTDV) program. Testing the combustor for a "long duration" is a massive technical challenge because the air flows through the engine in milliseconds, making it incredibly difficult to sustain a stable flame and generate continuous thrust at those extreme speeds.
- These typically rely on solid-propellant rocket motors (like the Nag or Astra missiles) because they require immediate, explosive acceleration over relatively short distances. They do not operate long enough or fast enough to require air-breathing scramjet engines.
- Naval vessels are powered by gas turbines, diesel engines, or nuclear reactors, none of which utilize supersonic air-breathing propulsion systems.

Q 16. Ans. D

Explanation:

- Amoeba reproduces by binary fission, in which the parent cell divides into two equal daughter cells.
- Hydra and yeast reproduce by budding, where a small outgrowth develops and detaches as a new individual.
- Spirogyra reproduces by fragmentation, in which the filament breaks into pieces, each growing into a new organism.

Q 17. Ans. B

Explanation:

- **Pair 1 is correctly matched:** A **dominant trait** is an inherited characteristic that appears in an offspring even if only one allele for that trait is inherited from a parent. For example, in pea plants, the allele for tallness (T) is dominant, so both TT and Tt genotypes will result in a tall plant.
- **Pair 2 is correctly matched:** A **recessive trait** is a trait that is expressed only when an organism has two recessive alleles for that trait (a homozygous state). The dominant allele must be absent. For example, a pea plant will only be short if its genotype is tt.

- **Pair 3 is incorrectly matched: Homologous organs** have the same basic structural design and origin but perform different functions. They are a sign of divergent evolution. An example is the forelimb of a human (used for grasping) and the flipper of a whale (used for swimming).
- **Pair 4 is incorrectly matched: Analogous organs** have different basic anatomical structures and origins but perform similar functions. They indicate convergent evolution. An example is the wings of birds (made of bone and feathers) and the wings of insects (made of membranes).

Q 18. Ans. B

Explanation:

- In 1829, the German chemist **Johann Döbereiner** observed that certain elements with similar chemical properties could be grouped together in clusters of three, which he called Triads.
- **His most significant observation within these triads was that the atomic mass of the middle element was roughly the average of the atomic masses of the other two elements. A classic example is the alkali metal triad: Lithium (Li), Sodium (Na), and Potassium (K).**
- **John Newlands:** Proposed the Law of Octaves in 1865. He arranged the known elements in order of increasing atomic mass and noted that every eighth element shared similar properties, much like the musical octaves.
- **Dmitri Mendeleev:** Often called the father of the periodic table, he formulated the Periodic Law in 1869, arranging elements by atomic mass and chemical properties. Crucially, he left empty gaps for undiscovered elements and successfully predicted their properties.
- **Henry Moseley:** In 1913, he revolutionized the classification system by proving that the fundamental property of an element is its atomic number (number of protons), not its atomic mass, leading to the Modern Periodic Table we use today.

Q 19. Ans. A

Explanation:

- **Assertion (A) is true:** Quantum information scrambling is a process in which quantum information is not destroyed but becomes highly delocalized across an entangled many-body quantum system. This is consistent with the principles of quantum mechanics, which preserve information during unitary evolution.

- **Reason (R) is true:** During scrambling, information initially encoded in a single qubit spreads across many qubits through entanglement. As a result, the original information cannot be recovered through measurements on any small subset of qubits and requires access to a significant portion of the system.
- **Reason (R) correctly explains Assertion (A):** Information remains preserved because it is redistributed throughout the entangled quantum system rather than erased. The dispersal of information among multiple qubits is precisely what is meant by quantum information scrambling.

Q 20. Ans. B

Explanation:

- The **25th Shanghai Cooperation Organisation (SCO) Summit**, held in **Tianjin, China, in 2025**, approved the **SCO Development Strategy until 2035** as one of its key outcome documents. The strategy outlines the organisation's long-term vision for cooperation in areas such as security, economic development, connectivity, and regional stability.

Q 21. Ans. D

Explanation:

- India's Biennial Update Report (BUR) is submitted to the United Nations Framework Convention on Climate Change (UNFCCC) as part of the reporting requirements for developing countries under the Convention. The report provides updated information on national greenhouse gas inventories, mitigation actions, and support needs related to climate change.
- India submitted its Fourth Biennial Update Report (BUR-4) to the UNFCCC on 30 December 2024. The report includes the national greenhouse gas inventory for 2020, details of mitigation measures, and progress towards climate-related commitments.

Q 22. Ans. B

Explanation:

- **Statement 1 is correct:** The Foreign Currency Settlement System (FCSS) enables foreign currency transactions between IFSC Banking Units (IBUs) operating in GIFT IFSC to be settled locally, eliminating the need for routing such transactions through overseas correspondent banks.
- **Statement 2 is correct:** The FCSS has been introduced under the regulatory framework of the **International Financial Services Centres Authority**

(IFSCA) to strengthen the financial ecosystem of GIFT IFSC and facilitate efficient cross-border transactions.

- **Statement 3 is incorrect:** The FCSS is not limited to settlement in euros alone. It has been designed to support settlement in multiple major foreign currencies, subject to operational and regulatory arrangements.
- **Statement 4 is correct:** One of the key objectives of the FCSS is to reduce dependence on the traditional correspondent banking route, thereby lowering transaction costs, improving settlement efficiency, and reducing operational risks associated with cross-border payments.

Q 23. Ans. D

Explanation:

- **Statement 1 is correct:** The FRBM Act, 2003 requires the Central Government to place before Parliament a **Medium-Term Fiscal Policy Statement**, along with other fiscal policy statements, to ensure transparency and accountability in fiscal management.
- **Statement 2 is correct:** The original framework of the FRBM Act envisaged reducing the fiscal deficit to 3% of GDP and eliminating the revenue deficit in a phased manner to achieve fiscal discipline.
- **Statement 3 is correct:** Based on the recommendations of the N.K. Singh Committee, the **Finance Act, 2018** amended the FRBM framework and shifted the focus towards a debt-based fiscal anchor. Consequently, the statutory targets relating to **Revenue Deficit** and **Effective Revenue Deficit** were removed from the Act.

Q 24. Ans. A

Explanation:

- **Statement A is incorrect:** The Hollongapar Gibbon Wildlife Sanctuary is the only wildlife sanctuary in India named after an ape species, the Hoolock Gibbon, but it is not the only protected area in India named after an ape species.
- **Statement B is correct:** The sanctuary is primarily known for the conservation of the Western Hoolock Gibbon (Hoolock hoolock), India's only ape species and the only ape found in the country.
- **Statement C is correct:** Hollongapar Gibbon Wildlife Sanctuary preserves one of the last remaining patches of tropical semi-evergreen and evergreen forests in the Brahmaputra Valley, making it an important biodiversity refuge in Assam.

- **Statement D is correct:** The sanctuary is located in Jorhat district of Assam, on the southern side of the Brahmaputra River, between the rivers Dissoi and Bhogdoi.

Q 25. Ans. A

Explanation:

- **Statement 1 is correct:** Gene Drive Technology (GDT) promotes the inheritance of a particular gene at rates significantly higher than the 50% probability expected under normal Mendelian inheritance. As a result, the targeted gene can spread rapidly through a population over successive generations.
- **Statement 2 is correct:** Gene drives have been proposed as a tool to control populations of disease-carrying mosquitoes, such as those transmitting malaria, dengue, and Zika, as well as certain agricultural pests that cause crop losses.
- **Statement 3 is incorrect:** Gene Drive Technology does not necessarily require the insertion of foreign DNA into the target organism. A gene drive can modify or propagate existing genes within a species, and the technology's defining feature is biased inheritance rather than the introduction of foreign genetic material.

Q 26. Ans. A

Explanation:

- **Statement 1 is correct:** Gram Kachahari is a distinctive feature of Bihar's Panchayati Raj system. Established under the **Bihar Panchayat Raj Act, 2006**, it serves as a village-level institution for the resolution of minor disputes. Its objective is to provide **accessible, inexpensive, and speedy justice** at the grassroots level and reduce the burden on regular courts.
- **Statement 2 is correct:** The **Sarpanch** is the elected head of the Gram Kachahari and is chosen directly by the voters of the Gram Panchayat area. The Sarpanch presides over proceedings and plays a key role in dispute resolution.
 - ✓ **Mukhiya** → Head of Gram Panchayat (executive body)
 - ✓ **Sarpanch** → Head of Gram Kachahari (judicial body)
- **Statement 3 is incorrect:** Gram Kachahari does **not** have jurisdiction over all criminal offences. Its powers are restricted to certain **minor offences and petty disputes** specifically notified under law. Serious offences such as: Murder, Dacoity, Rape, Kidnapping, Major offences under the IPC remain under the jurisdiction of regular criminal courts.

Q 27. Ans. B

Explanation:

- **Statement 1 is correct:** The Patna High Court began functioning on **1 March 1916**. At the time of its establishment, the Court consisted of **one Chief Justice and six Puisne Judges** (ordinary judges other than the Chief Justice). This was the original sanctioned strength prescribed under the Letters Patent establishing the Court.
- **Statement 2 is correct:** **Sir Edward Maynard Des Champs Chamier** was appointed as the **first Chief Justice of the Patna High Court** when it commenced functioning in 1916. He played a significant role in organizing the newly established High Court and laying the foundation of its judicial administration.
- **Statement 3 is incorrect:** Although **Orissa became a separate province on 1 April 1936**, the Patna High Court **continued to exercise jurisdiction over Orissa** even after separation.
- A separate **Orissa High Court** was established only on **26 July 1948**. Until then, cases from Orissa continued to come under the jurisdiction of the Patna High Court.
- **Statement 4 is correct:** The Letters Patent creating the Patna High Court provided for **circuit sittings at Cuttack**, since Orissa was then part of the jurisdiction of the Patna High Court. This arrangement ensured judicial accessibility for litigants from the Orissa region before the establishment of a separate High Court.

Q 28. Ans. C

Explanation:

- According to the **Economic Survey of Bihar 2024–25** and **Budget Analysis 2026–27**, Bihar's **per capita Gross State Domestic Product (GSDP) at current prices** is estimated at **₹76,490 in 2024–25**, registering a growth of approximately **13%** over the previous year. Despite recording a higher growth rate in per capita income than the national average (**13% vs 9%**), Bihar continues to have one of the **lowest per capita incomes among Indian states**. Bihar's per capita GSDP is approximately **one-third of the national per capita GDP**,

Q 29. Ans. C

Explanation:

- **Patna College** was established on **9 January 1863** by the British colonial government as a government institution to cater to the growing demand for

higher education in the region. Initially, the college functioned from buildings that had once been part of a 17th-century Dutch factory. Later, Patna University was established on **1 October 1917**, becoming the first university in Bihar and the seventh oldest university in the modern Indian subcontinent.

- **Bihar Agricultural University (BAU)**, located at Sabour in Bhagalpur district, is an autonomous state agricultural university dedicated to education, research, and extension activities in agriculture and allied sciences. Its historical roots trace back to 1908, when Bihar Agricultural College was established as the Bengal Provincial Agricultural College by Andrew Henderson Leith Fraser. Following the partition of Bengal, the institution was renamed Bihar Agricultural College. Subsequently, Bihar Agricultural University was formally established on **5 August 2010** by an Act of the Bihar State Legislature.
- **Jai Prakash University (JPU)** was established on **22 November 1990** by the Government of Bihar under Section 3(b) of the Bihar State Universities Act, 1976. The university was carved out of B.R.A. Bihar University (formerly Bihar University) to meet the growing higher education needs of the Saran Division. Headquartered in Chapra, the university was created to serve the districts of Saran, Siwan, and Gopalganj.

Q 30. Ans. D

Explanation:

- **Statement 1 is correct:** The **37th annual session of the Indian National Congress (INC)** was held at **Gaya** from **26 to 31 December 1922** under the presidency of **Chittaranjan Das**. A major issue debated during the session was the future strategy of the national movement after the suspension of the Non-Cooperation Movement. Chittaranjan Das, along with **Motilal Nehru**, advocated ending the boycott of legislative councils and argued that nationalists should contest elections, enter the legislative assemblies, and obstruct British policies from within the colonial system. However, their proposal was rejected by the Congress leadership. Following this defeat, Chittaranjan Das resigned from the Congress presidency and, along with Motilal Nehru, founded the **Congress-Khilafat Swaraj Party** (commonly known as the **Swaraj Party**) on **1 January 1923** to participate in legislative council elections and pursue constitutional opposition to British rule.

- **Statement 2 is correct:** The Prince of Wales visited Patna on 22 December 1921, arriving from Nepal by steamer and train and landing at Commissioner's Ghat. He was received with a semi-state procession that proceeded to **Gandhi Maidan** (then known as Bankipore Maidan), where a grand reception durbar was organized in his honor. The welcome address was delivered by the eminent lawyer and freedom fighter **Sachchidanand Sinha**. To commemorate this royal visit, the **Prince of Wales Medical College** was established in 1925, which later evolved into **Patna Medical College and Hospital (PMCH)**, one of the oldest and most prestigious medical institutions in eastern India.

Q 31. Ans. A

Explanation:

- **Statement 1 is correct:** The Charter Act of 1833 made Lord William Bentinck the first Governor General of India, vesting in him exclusive legislative powers and superintendence over the Bombay and Madras presidencies. It was the final step towards the centralisation of administration in British India.
- **Statement 2 is incorrect:** Open competition for civil services was provided by the Charter Act of 1853, which threw the covenanted civil service open to Indians and ended the patronage of the Court of Directors. The 1833 Act had attempted a merit principle but it was negated by the Court of Directors.
- **The Charter Acts in sequence:** the Charter Act of 1793 renewed the Company's commercial privileges; the Charter Act of 1813 ended the Company's trade monopoly except in tea and trade with China; the Charter Act of 1833 ended even the China and tea monopoly and made the Company a purely administrative body holding territories in trust for the Crown; and the Charter Act of 1853 was the last in the series and separated the legislative and executive functions of the Governor General's council.

Q 32. Ans. A

Explanation:

- The Union Powers Committee was chaired by Jawaharlal Nehru, who also chaired the Union Constitution Committee and the States Committee dealing with negotiations with the princely states.
- The Drafting Committee was chaired by B. R. Ambedkar and had seven members, including N. Gopalaswami Ayyangar, Alladi Krishnaswami

Ayyar, K. M. Munshi, Mohammad Saadulla, N. Madhava Rau (replacing B. L. Mitter), and T. T. Krishnamachari (replacing D. P. Khaitan).

- The Provincial Constitution Committee was chaired by Sardar Vallabhbhai Patel, who also chaired the Advisory Committee on Fundamental Rights, Minorities and Tribal and Excluded Areas.
- The Rules of Procedure Committee was chaired by Rajendra Prasad, who was also the permanent President of the Constituent Assembly and chaired the Steering Committee. The Assembly was set up under the Cabinet Mission Plan of 1946, held its first meeting on 9 December 1946 with Sachidananda Sinha as temporary chairman, adopted the Objectives Resolution moved by Nehru, and took two years, eleven months and eighteen days to complete the Constitution, which was adopted on 26 November 1949 and enforced on 26 January 1950.

Q 33. Ans. C

Explanation:

- **Assertion is true:** In the Kesavananda Bharati case of 1973, the Supreme Court held that the Preamble is a part of the Constitution and can be amended under Article 368, subject to the basic structure. This was reaffirmed in the LIC of India case of 1995.
- **Reason is false:** In the Berubari Union case of 1960, the Supreme Court held that the Preamble is not a part of the Constitution, treating it merely as a key to the minds of the makers. The Kesavananda Bharati case later reversed this specific position.
- The Preamble has been amended only once, by the 42nd Amendment Act of 1976, which added the words socialist, secular, and integrity. The Preamble is non-justiciable and is neither a source of legislative power nor a limitation upon it; its key words are sovereign, socialist, secular, democratic, and republican, and it declares the date of adoption as 26 November 1949.

Q 34. Ans. A

Explanation:

- **Statement 1 is correct:** A bill under Article 3 can be introduced in either House of Parliament only on the prior recommendation of the President. Before recommending, the President must refer the bill to the concerned state legislature for its views within a specified period.
- **Statement 2 is incorrect:** There is no requirement of ratification by the state legislatures, and the

views expressed by the state legislature are not binding on Parliament, which may accept or reject them. A law made under Article 3 is not treated as a constitutional amendment under Article 168 and is passed by a simple majority.

- Article 1 describes India as a Union of States, while Article 2 deals with the admission or establishment of new states and Article 3 with the internal reorganisation of existing states. In the *Berubari Union case*, the Court held that ceding Indian territory to a foreign state requires a constitutional amendment under Article 368, not a law under Article 3. The 100th Amendment Act of 2015 gave effect to the land boundary agreement with Bangladesh through the exchange of enclaves.

Q 35. Ans. A

Explanation:

- **Statement 1 is correct:** Article 20(1) protects against ex post facto criminal laws by barring conviction for an act that was not an offence when committed, and barring a penalty greater than that prescribed at the time. This protection covers only criminal laws, not civil laws or tax laws, and does not extend to retrospective changes in procedure or to preventive detention.
- **Statement 2 is correct:** Articles 20 and 21 cannot be suspended even during a national emergency. This safeguard was made absolute by the 44th Amendment Act of 1978, which amended Article 359 in response to the position taken in the *ADM Jabalpur case* during the 1975 emergency.
- **Statement 3 is incorrect:** Article 20 is available to all persons, including citizens, foreigners, and legal persons such as companies and corporations. Article 20 also contains the protection against double jeopardy under clause (2), which is narrower than the American doctrine because it bars a second prosecution and punishment only where there was both prosecution and punishment in a prior judicial proceeding, and the protection against self incrimination under clause (3), which the Court in the *Kathi Kalu Oghad case* held does not cover specimen signatures, thumb impressions, or blood samples.

Q 36. Ans. A

Explanation:

- **Statement 1 is correct:** The Lower Palaeolithic Age represents the earliest phase of the Stone Age in the Indian subcontinent and is associated

primarily with the Acheulian tool tradition. The most characteristic and diagnostic tools of this period were hand-axes and cleavers, which were manufactured mainly from quartzite and other hard stones.

- ✓ Hand-axes were bifacially worked tools, usually almond-shaped, pear-shaped, or ovate in form.
- ✓ Cleavers were large cutting tools distinguished by a broad transverse cutting edge.
- These tools were generally used for hunting, butchering animals, chopping wood, and other subsistence activities.
- The prevalence of hand-axes and cleavers is so significant that the Lower Palaeolithic culture in India is often referred to as the Acheulian Culture.
- **Statement 2 is incorrect:** Lower Palaeolithic sites were not confined exclusively to the peninsular river valleys. Although many important sites have been discovered in river valleys such as the Narmada, Godavari, Krishna, and Tungabhadra basins, the geographical distribution of Lower Palaeolithic cultures was much wider.
 - ✓ Evidence has been found in northwestern regions, particularly the Soan Valley.
 - ✓ Important sites occur in the Belan Valley of Uttar Pradesh, showing the presence of Lower Palaeolithic populations beyond peninsular India.
 - ✓ Discoveries from Rajasthan, Madhya Pradesh, Uttar Pradesh, and other regions demonstrate that this culture was spread across large parts of the Indian subcontinent.

Q 37. Ans. B

Explanation:

- Burzahom is one of the most important Neolithic sites in the Indian subcontinent, located in the Kashmir Valley of present-day Jammu and Kashmir. Excavations at the site have revealed several distinctive cultural features that set it apart from other Neolithic settlements.
- **Burzahom is famous for:**
 - ✓ Evidence of Pit-Dwellings
 - The most characteristic feature of Burzahom is its pit-dwelling habitations.
 - Early inhabitants dug large circular or oval pits into the ground and used them as residential structures, particularly during the harsh winters of the Kashmir Valley.
 - ✓ Human Burials with Dogs

- Burzahom has yielded unique burial practices, including instances where dogs were buried alongside their masters.
- Such burials indicate the close relationship between humans and domesticated animals and provide valuable insights into the social and cultural life of Neolithic communities.

Q 38. Ans. A

Explanation:

- **Statement 1 is correct:** Navdatoli, situated on the banks of the Narmada River in Madhya Pradesh, is one of the most significant Chalcolithic sites associated with the Malwa Culture. Archaeological excavations have revealed an exceptionally diverse range of cultivated crops, making it one of the richest sources of evidence for Chalcolithic agriculture in the Indian subcontinent.
- **The site has yielded remains of:**
 - ✓ Wheat
 - ✓ Barley
 - ✓ Rice
 - ✓ Lentil
 - ✓ Green gram (moong)
 - ✓ Pea
 - ✓ Linseed
 - ✓ **Other pulses and food crops**
- **Statement 2 is incorrect:** The Chalcolithic cultures of Eastern India, particularly in regions of present-day West Bengal and Bihar, followed a subsistence pattern that differed from that of western and central India. Eastern Chalcolithic communities heavily supplemented their diet with fish as evidenced by the discovery of copper fish hooks and fish bones at sites like Pandu Rajar Dhibi.
 - ✓ The principal crop cultivated in these settlements was rice, owing to the region's favourable climatic and ecological conditions.
 - ✓ Sites such as Chirand, Mahisdal, and Pandu Rajar Dhibi have produced substantial evidence of rice cultivation.

Q 39. Ans. D

Explanation:

- **Statement 1 is correct:** The Rig Vedic economy (c. 1500–1000 BCE) was predominantly pastoral in character, although agriculture was also known. Wealth and social status were measured primarily in terms of cattle ownership, making cows the most important economic asset of the period.

- ✓ The Rig Veda contains numerous references to cattle wealth, cattle raids (gavishthi), and gifts of cows.
- ✓ Cows served as the principal indicator of prosperity and were often used as a medium for exchange and valuation.
- ✓ The term gomati (possessing cows) was synonymous with being wealthy.
- ✓ Wars between tribes were frequently fought for the acquisition of cattle, highlighting their economic significance.
- ✓ Although horses, sheep, and goats were also reared, cattle constituted the core of the pastoral economy.

- **Statement 2 is incorrect:** The Rig Vedic economy had not yet developed a system of coinage. Trade and exchange were conducted mainly through:

- ✓ Barter transactions
- ✓ Exchange of cattle
- ✓ Gifts and tributes.

- Nishka was generally a gold ornament or gold piece worn around the neck, and later came to be used as a unit of value, but it did not functioned as a standardised coin in the Rig Vedic period.
- The use of metallic coins in India emerged much later, with the appearance of Punch-Marked Coins during the post-Vedic period (around the 6th century BCE).

Q 40. Ans. B

Explanation:

- The Rig Vedic polity was essentially tribal and monarchical, though the king (Rajan) was not an absolute ruler. His authority was limited by tribal customs and by popular assemblies such as the Sabha, Samiti, Vidatha, and Gana. To assist him in the administration and management of tribal affairs, several important officials existed.
 - ✓ **Purohita (Chief Priest)**
 - The Purohita was the most influential adviser of the king.
 - He performed religious rituals and sacrifices on behalf of the tribe.
 - He was believed to ensure divine protection and success in warfare through prayers and sacrifices.
 - Prominent Rig Vedic priests such as Vashistha and Vishvamitra enjoyed considerable political influence.
 - ✓ **Senani (Military Commander)**
 - The Senani was the chief military officer of the tribe.

- He organised and led the tribal army during conflicts and cattle raids.
- Since inter-tribal warfare was common, the Senani occupied an important position in the political structure.
- ✓ **Vrajapati**
 - The Vrajapati was associated with the supervision and management of grazing grounds, cattle herds, and pastoral resources.
 - Given the predominantly pastoral nature of the Rig Vedic economy, this office held considerable significance.

Q 41. Ans. B

Explanation:

- The doctrines of Anekantavada and Syadvada are among the most distinctive philosophical contributions of Jainism. These concepts were developed to explain the complex nature of reality and the limitations of human knowledge.
 - ✓ The term Anekantavada literally means “the doctrine of many-sidedness”.
 - ✓ According to Jain philosophy, reality is complex and possesses multiple aspects.
- No single statement or viewpoint can fully describe the entire truth.
 - ✓ Different perspectives may reveal different aspects of the same reality, and each may contain a portion of the truth.
- This doctrine promotes intellectual tolerance and rejects absolute or dogmatic assertions.
- **Syadvada (Doctrine of conditioned predication):** This is the practical application of Anekantavada. Because reality is multifaceted, Jainism argues that all human judgments and statements are conditional and relative. Therefore, every statement should ideally be prefaced with “Syat” (meaning “from a certain perspective” or “maybe”). This prevents dogmatism and is mathematically expressed in the seven-fold formula known as Saptabhangi Naya.

Q 42. Ans. C

Explanation:

- **Statement 1 is correct:** The Arthashastra, traditionally attributed to Kautilya (also known as Chanakya or Vishnugupta), presents the famous Saptanga Theory of State. According to this theory, a state is composed of seven essential limbs (Saptangas):
 - ✓ Swami (King)

- ✓ Amatya (Ministers and officials)
- ✓ Janapada (Territory and population)
- ✓ Durga (Fortified capital)
- ✓ Kosha (Treasury)
- ✓ Danda/Bala (Army)
- ✓ Mitra (Allies)

- Kautilya compared the state to a living organism in which all limbs are interdependent. The efficiency and stability of the state depended upon the proper functioning of each component. A weakness in any one limb could adversely affect the strength and effectiveness of the entire state.
- **Statement 2 is correct:** The Arthashastra describes a highly developed and centralized administrative system with numerous specialized departments. Each department was headed by a superintendent known as an Adhyaksha, who was responsible for regulating and supervising a specific sphere of administration and economic activity.
- **Examples include:**
 - ✓ Akaradhyaksha – Superintendent of Mines
 - ✓ Panyadhyaksha – Superintendent of Trade and Commerce
 - ✓ Sitadhyaksha – Superintendent of Agriculture
 - ✓ Sulkadhyaksha – Superintendent of Customs and Tolls
 - ✓ Navadhyaksha – Superintendent of Shipping and Navigation

Q 43. Ans. C

Explanation:

- **Statement 1 is correct:** One of the major economic developments of the post-Gupta period (c. 600–1200 CE) was the gradual decline of long-distance trade, particularly the flourishing Indo-Roman commerce that had enriched many urban centres during the early centuries of the Common Era.
 - ✓ During the Kushana and Gupta periods, Indian merchants maintained extensive trade links with the Roman Empire, exporting spices, textiles, precious stones, ivory, and luxury goods.
 - ✓ From the 3rd century CE onward, the decline of Roman economic power and changing international trade patterns reduced the volume of this commerce.
 - ✓ As trade contracted, many urban centres that had depended on commercial activity experienced stagnation or decline.
- **Statement 2 is correct:** Another important feature of the post-Gupta economy was the growing practice

of land grants by kings to Brahmanas, religious institutions, temples, and administrative officials.

- ✓ These grants often transferred not only land but also various revenue and administrative rights.
- ✓ Many charters exempted the grantees from royal taxes and granted them rights over local resources, fines, and dues.
- ✓ Such grants frequently carried fiscal and administrative immunities, meaning royal officials could not interfere in the granted territories.
- ✓ Over time, the beneficiaries acquired substantial economic and social power over the local population.
- ✓ Historians view this process as contributing to the emergence of a landed intermediary class, often described within the framework of Indian feudalism.

Q 44. Ans. C

Explanation:

- **Bhimbandh Wildlife Sanctuary** is located in the south-western part of **Munger district** across the hills and undulating terrain of the **Kharagpur Hills**. It is situated **south of the Ganga River**, along the **northern edge of the Chota Nagpur Plateau** and to the **west of the Santhal Pargana region**. The sanctuary is also renowned for its natural hot springs located in the valleys and foothill areas, the most notable being **Bhimbandh, Sita Kund, and Rishi Kund**.
- **Baraila Lake Salim Ali-Jubba Sahni Bird Sanctuary**, popularly known as **Baraila Bird Sanctuary**, is a protected wildlife sanctuary located in **Vaishali district**. The lake is associated with local traditions linked to the **Ramayana**. According to popular legend, **Rama** halted here while traveling to **Janakpur** (Mithila) for his marriage. Delighted by his arrival, the local people are said to have exclaimed “**Var Aila**” (meaning “the groom has arrived”), a phrase that gradually evolved into the name “**Baraila**.”
- **Kusheshwar Asthan Bird Sanctuary** is a protected wetland and bird sanctuary located in **Darbhanga district**. The sanctuary derives its name from the nearby **Kusheshwar Asthan Temple**, a renowned Hindu pilgrimage site dedicated to **Lord Shiva**.
- **Udaypur Wildlife Sanctuary** (also spelled **Udaipur Wildlife Sanctuary**) is located in **West Champaran district**. Established in **1978**, the sanctuary

covers an area of approximately **8.74 sq. km**. It is predominantly a wetland ecosystem centered around an **oxbow lake** formed in the floodplain of the **Gandak River**. Its vegetation includes **swamp forests, dry riverine forests, and khair-sissoo forests** dominated by **Acacia catechu** and **Dalbergia sissoo**.

Q 45. Ans. C

Explanation:

- **Pair 1 is incorrect:** Founded in **Patna on 15 July 1918**, **The Searchlight** is regarded as the **first English daily newspaper of the province**. Associated with prominent nationalist leaders such as **Dr. Sachchidananda Sinha**.
- **Pair 2 is incorrect:** The **Bihar Herald (Behar Herald)**, established in **1874/75** at **Patna**, is regarded as the **first English newspaper and weekly of Bihar**. It emerged during a period when educated Biharis were demanding a separate administrative identity for Bihar from the Bengal Presidency.
- **Pair 3 is incorrect:** **Bihar Bandhu** was launched in **1872/74** and is widely recognized as the **first Hindi newspaper of Bihar**. Published as a weekly, it played a pioneering role in the promotion of the Hindi language, literature, and public awareness during the colonial period.
- **Pair 4 is incorrect:** **Sarvhitashi** is regarded as the **first Hindi daily newspaper published in Bihar**. The newspaper contributed to the growth of Hindi journalism and helped expand the reach of daily news among Hindi readers in the province.

Q 46. Ans. A

Explanation:

- **Statement 1 is correct:** Geographical Indications (GI tags) are recognized as a form of **Intellectual Property Right (IPR)**. They are used to identify products that originate from a specific geographical region and possess distinctive qualities, reputation, or characteristics attributable to that place of origin. GI registration provides legal protection against unauthorized use and helps preserve traditional products and cultural heritage.
- **Statement 2 is correct:** The **Geographical Indications Registry** is responsible for the registration and administration of GI tags in India. It functions under the **Department for Promotion of Industry and Internal Trade (DPIIT)**, which is a department of the **Ministry of Commerce and Industry, Government of India**. The Registry

implements the provisions of the **Geographical Indications of Goods (Registration and Protection) Act, 1999**.

- **Statement 3 is incorrect:** A registered GI does not lapse permanently after twenty years. Under the **Geographical Indications of Goods (Registration and Protection) Act, 1999**, a GI registration is valid for an **initial period of ten years** and can be **renewed indefinitely** for successive ten-year periods upon payment of the prescribed fee. Therefore, there is no maximum time limit of twenty years for GI protection.
- **Statement 4 is incorrect:** The **Geographical Indications Registry** is located in **Chennai, Tamil Nadu**, not in Delhi. It operates under the office of the **Controller General of Patents, Designs and Trade Marks (CGPDTM)** and serves as the central authority for GI registration in India.

Q 47. Ans. A

Explanation:

- **Statement 1 is incorrect:** JEEViKA, formally known as the **Bihar Rural Livelihoods Promotion Society (BRLPS)**, was launched by the Government of Bihar on **2 October 2006** and constituted under the **Societies Registration Act, 1860**.
- **Statement 2 is correct:** JEEViKA, formally known as the **Bihar Rural Livelihoods Promotion Society (BRLPS)** started by the Government of Bihar with the financial assistance of World Bank launched an ambitious poverty alleviation project. Following the launch of the **National Rural Livelihoods Mission** in December 2010, the Bihar Government adopted the **JEEViKA Model** for statewide implementation and designated JEEViKA as the **State Rural Livelihood Mission**, paving the way for its phased expansion across all blocks of Bihar.
- **Statement 3 is incorrect:** JEEViKA began as a pilot project in five villages across five districts—**Muzaffarpur, Nalanda, Madhubani, Purnia, and Gaya**. Encouraged by its initial success, the programme was gradually expanded to additional blocks and districts, including Khagaria, Madhepura, Supaul, and Saharsa, with some areas being covered under the Bihar Kosi Flood Recovery Project.

Q 48. Ans. D

Explanation:

- **Statement 1 is correct:** The **Phalgu (Falgu) River**, one of the most sacred rivers of Bihar, is formed

by the confluence of the **Lilajan (Niranjana)** and **Mohana** rivers near **Gaya**. After flowing past Gaya, it ultimately joins the **Punpun River**. The river holds immense religious significance for both Hindus and Buddhists and is closely associated with the performance of **Pind Daan** rituals at Gaya.

- **Statement 2 is correct:** The **Phalgu** is historically identified with the **Niranjana River** mentioned in Buddhist texts. It was on the banks of this river that **Gautama Buddha** practiced severe austerities before realizing the Middle Path.
- **Statement 3 is correct:** The **Sujata Stupa** is situated at **Bakraur village**, directly across the **Phalgu River** from Bodh Gaya. The stupa commemorates **Sujata**, whose offering of milk-rice to **Siddhartha Gautama** marked a turning point in his spiritual journey and the adoption of the Middle Way.
- **Statement 4 is correct:** the river is also popularly known as the **“Hidden Ganga” (Gupta Ganga)** because for much of the year it appears as a vast sandy bed with water flowing beneath the surface.

Q 49. Ans. C

Explanation:

- **Statement 1 is correct:** Quantum Information Scrambling refers to the process by which initially localized quantum information spreads rapidly across a many-body quantum system through interactions and entanglement. As a result, the information becomes encoded in complex correlations involving a large number of particles rather than remaining confined to a specific location.
- **Statement 2 is correct:** During scrambling, the information is not destroyed; instead, it becomes inaccessible to local observations or measurements. Recovering the original information generally requires access to a substantial part of the entire quantum system, reflecting the non-local nature of quantum correlations.
- **Statement 3 is correct:** Recent experiments have shown that quantum information scrambling can be used to demonstrate a verifiable quantum advantage. Certain scrambling-related tasks can be efficiently performed and verified using quantum processors but are extremely difficult for classical computers to simulate, highlighting the computational power of quantum systems.
- **Statement 4 is incorrect:** Quantum information scrambling does not imply the permanent loss of quantum information. According to the principles

of quantum mechanics, particularly unitary evolution, the information remains preserved within the global quantum state. It is effectively hidden or dispersed, but not erased.

Q 50. Ans. C

Explanation:

- The Reserve Tranche Position (RTP) represents the portion of a member country's quota that remains readily available for withdrawal from the International Monetary Fund (IMF). It is essentially the difference between a member's quota subscription and the IMF's holdings of that member's currency that are not part of IMF lending operations.
- The RTP constitutes a member country's reserve asset with the IMF and can be accessed without conditionality or the need to negotiate a loan programme. It arises when the IMF uses a member country's currency to provide financial assistance to other member countries.

Q 51. Ans. A

Explanation:

- **Assertion (A) is true** North Bihar, being a part of the vast **Middle Gangetic Plain**, experiences dense fog and prolonged foggy conditions during the winter season (December–January). Districts such as **Sitamarhi, Madhubani, Darbhanga, Supaul, Araria, Purnia** etc.
- **Reason (R) is true:** Temperature inversion commonly develops over the Gangetic Plain during winter when the ground cools rapidly at night, causing cold air to remain trapped near the surface beneath a layer of warmer air. This prevents vertical mixing and promotes the formation of dense fog.

Q 52. Ans. C

Explanation:

- **Statement 1 is correct:** Diara lands are low-lying alluvial tracts formed along the banks and floodplains of major rivers such as the **Ganga, Gandak, Kosi, Ghaghara, Son, and Mahananda**. These lands are created through the continuous deposition of sediments carried by rivers during floods. During the monsoon season, rivers overflow their banks and deposit layers of sand, silt, clay, and fine alluvium. Over time, these deposits accumulate and form new land surfaces known as **Diara lands**.
- **Statement 2 is incorrect:** Diara lands are located in

active floodplain regions where rivers frequently change their course through erosion and deposition. Therefore, their boundaries are dynamic and do **not remain stable over long periods**.

- **Statement 3 is correct:** Since Diara lands receive fresh sediment deposits during recurring floods, they are generally composed of recent alluvium (**Khadar**) and contain some of the youngest and most fertile alluvial soils found in Bihar.

Q 53. Ans. D

Explanation:

- **Murli Bharahwa village in present-day West Champaran** is associated with **Raj Kumar Shukla**, the indigo cultivator who played a pivotal role in bringing Mahatma Gandhi to Champaran. Distressed by the exploitation of peasants under the **Tinkathia system**, Shukla persistently persuaded Gandhi to visit Champaran. His efforts ultimately led to the **Champaran Satyagraha (1917)**, Gandhi's first successful Satyagraha in India.
- **Sadaqat Ashram**, located on the banks of the Ganga in Patna, was established by **Mazhar-ul-Haq** during the Non-Cooperation Movement. It became an important centre of nationalist activities in Bihar and served as a meeting place for freedom fighters. After demitting office as the first President of India, **Dr. Rajendra Prasad** spent his final years here. Today, it houses the **Bihar Vidyapeeth** and the **Rajendra Smriti Sangrahalaya**.
- **Koeldih (Koel Deeh)** in Champaran is associated with Gandhi's visit during the Champaran Satyagraha. It was one of the villages where Gandhi interacted directly with indigo cultivators and recorded their testimonies regarding exploitation by European planters. These firsthand accounts became crucial evidence before the Champaran Agrarian Committee and contributed to the abolition of the oppressive **Tinkathia system**.
- **Hardiya Kothi** was one of the prominent indigo factories (kothis) established by European planters in Champaran. Such kothis functioned as centres of indigo cultivation and administration. The planters compelled peasants to cultivate indigo on a fixed portion of their land under exploitative arrangements.

Q 54. Ans. C

Explanation:

- Bihar has increasingly focused on **solar energy development** because the state has historically faced

an energy deficit and possesses limited conventional energy resources within its present geographical boundaries. After the bifurcation of Bihar in 2000, most of the mineral-rich areas, including major coal reserves and several large thermal power assets, became part of Jharkhand. As a result, Bihar remained heavily dependent on electricity generated both within the state and procured from central generating stations and other states.

- To improve **energy security**, reduce dependence on fossil fuels, diversify its energy mix, and meet rising electricity demand, Bihar has promoted solar power through utility-scale solar parks, rooftop solar programmes, solar irrigation pumps, and decentralized renewable energy projects.

Q 55. Ans. D

Explanation:

- The nineteenth century witnessed several major military campaigns involving the British Empire. Although Britain ultimately emerged victorious in many of these conflicts, some wars exposed serious weaknesses and involved significant military reverses, defeats, or setbacks.
- **First Afghan War (1839–1842)**
 - ✓ British occupied Kabul and installed Shah Shuja.
 - ✓ Strong Afghan resistance soon emerged.
 - ✓ During the 1842 retreat from Kabul, almost the entire British force was annihilated.
 - ✓ Considered one of the worst military disasters in British imperial history.
- **Punjab Wars (1845–46 and 1848–49)**
 - ✓ Fought against the powerful and well-trained Sikh army.
 - ✓ The British suffered heavy casualties despite eventual victory.
 - ✓ The Battle of Chillianwala (1849) was a particularly severe setback.
 - ✓ Widely viewed as a major British military reverse.
- **Crimean War (1853–1856)**
 - ✓ Britain, France, the Ottoman Empire, and Sardinia together defeated Russia.
 - ✓ Exposed serious weaknesses in the British military organisation.
 - ✓ The Charge of the Light Brigade symbolised military mismanagement.
 - ✓ Poor logistics, disease, and heavy losses led to widespread criticism and reforms.

Q 56. Ans. A

Explanation:

- **Statement 1 is correct:** The Indian Councils Act of 1861 introduced only nomination, not election. The principle of election was introduced for the first time by the Indian Councils Act of 1892, though it was indirect and limited in nature. Direct elections came with the Morley-Minto Reforms of 1909.
 - ✓ Subsequent reforms, including the Indian Councils Acts of 1892 and 1909, enlarged the Council and permitted limited discussion of budgets and public matters.
 - ✓ Members could ask questions, debate certain issues, and make suggestions, but their powers remained highly restricted.
- **Statement 2 is incorrect:** The Imperial Legislative Council was not abolished by the Government of India Act of 1909. The 1909 Act (Morley-Minto Reforms) enlarged and reformed the Council but retained it. The Imperial Legislative Council was abolished by the Government of India Act of 1919 (Montagu-Chelmsford Reforms), which replaced it with a bicameral Central Legislature consisting of the Council of State and the Central Legislative Assembly.
 - ✓ The executive government remained responsible to the Governor-General and the British authorities, not to the Legislative Council.
 - ✓ Council members could not remove the executive through a vote of no confidence.
 - ✓ They had no authority to compel the executive to accept their recommendations.
 - ✓ Financial control was extremely limited, and large portions of the budget were beyond effective discussion or amendment.
 - ✓ The Governor-General retained extensive powers, including the power to certify bills and override opposition.
 - ✓ Even after the constitutional reforms of 1909 and 1919, executive accountability to the legislature remained largely absent.
 - ✓ The Council's role was largely consultative and advisory rather than decisive.

Q 57. Ans. B

Explanation:

- Lord Ripon is often regarded as the “Father of Local Self-Government in India” because of his famous Resolution on Local Self-Government issued in

1882. The objective of the scheme was to promote local administration and provide Indians with practical training in self-governance.

- **Under Ripon's local government reforms:**

- ✓ Local bodies such as municipalities and district boards were encouraged throughout British India.
- ✓ A larger number of non-official members were to be associated with these bodies.
- ✓ Wherever possible, non-official members were to be elected rather than nominated.
- ✓ However, the right to vote was not universal.

- **Voting rights were restricted on the basis of factors such as:**

- ✓ Property ownership
- ✓ Payment of taxes
- ✓ Income
- ✓ Social status and other qualifications prescribed by local regulations.

- As a result, only a small section of the population could participate in elections. The system was therefore based on a limited or restricted franchise, not on universal suffrage.

Q 58. Ans. D

Explanation:

- The construction of railways enabled British-manufactured goods to penetrate even the remotest villages of India.
- This accelerated the decline of traditional rural handicrafts and village industries, especially cotton spinning and weaving.
- D.H. Buchanan, an American writer, famously remarked that "armour of the isolated self-sufficient village was pierced by the steel rail, and its life blood ebbed away" in his landmark book, *The Development of Capitalistic Enterprise in India*.
- His statement highlighted how railways integrated villages into the colonial economy and undermined their traditional self-sufficient structure.

Q 59. Ans. D

Explanation:

- **Paika Rebellion (1817) — Bakshi Jagabandhu**

- ✓ Occurred in the Khurda region of Odisha in 1817.
- ✓ Caused by British revenue policies and loss of hereditary privileges of the Paikas (traditional militia).
- ✓ Led by Bakshi Jagabandhu, the military chief of the Khurda king.

- ✓ Rebels attacked British establishments and challenged colonial rule.

- **Ahom Revolt (1828–1830) — Gomdhar Konwar**

- ✓ Broke out in Assam after the British failed to restore Ahom rule as promised.
- ✓ Assamese nobles opposed continued British control.
- ✓ Gomdhar Konwar was proclaimed Ahom king and became the chief leader of the revolt.
- ✓ Though suppressed, the revolt forced the British to grant some concessions.

- **Kol Mutiny (1831–32) — Buddho Bhagat**

- ✓ Took place in the Chotanagpur region.
- ✓ Triggered by exploitation by moneylenders, zamindars, and British officials.
- ✓ Buddho Bhagat emerged as a prominent leader of the uprising.
- ✓ The revolt spread widely before being crushed.

- **Pagla Panthis — Tipu Shah**

- ✓ A semi-religious peasant movement in northern Bengal.
- ✓ Founded by Karam Shah and later led by his son Tipu Shah.
- ✓ Under Tipu Shah, it became strongly anti-zamindar and anti-British.
- ✓ Mobilised peasants and tribal communities against oppression.

Q 60. Ans. C

Explanation:

- **Rousseau — Democratic thought**

- ✓ The ideas of Jean-Jacques Rousseau, especially popular sovereignty and the social contract, influenced educated Indians.
- ✓ His writings encouraged democratic and representative political ideas.

- **Mazzini — Republican nationalism and nation-state ideology**

- ✓ Giuseppe Mazzini was the philosophical heart of Italian unification.
- ✓ His emphasis on secret societies ("Young Italy"), republican nationalism, and the ideological concept of a unified nation-state profoundly influenced Indian leaders.
- ✓ **V.D. Savarkar:** Modeled his secret society Abhinav Bharat directly after Mazzini's "Young Italy" and translated Mazzini's writings into Marathi.
- ✓ **Surendranath Banerjee:** Delivered highly popular lectures on Mazzini to the youth of Bengal to inspire patriotic devotion.

- **Garibaldi — Armed struggle for national liberation**

- ✓ Giuseppe Garibaldi was the military hero of the Italian Risorgimento.
- ✓ His leadership of the “Redshirts” and his successful armed guerrilla struggle for national liberation became the exact blueprint for Indian revolutionary terrorists who believed that ideological nationalism had to be backed by armed force.
- ✓ **Lala Lajpat Rai:** Wrote Urdu biographies of both Mazzini and Garibaldi, admiring both the ideological and military aspects of their struggle.
- **John Stuart Mill — Liberalism**
 - ✓ John Stuart Mill was associated with liberalism, individual liberty, and representative government.
 - ✓ He was not a socialist thinker in the sense implied by the question.
 - ✓ Socialist ideas in India were influenced more by thinkers such as Karl Marx and Friedrich Engels.

Q 61. Ans. A

Explanation:

- **Statement 1 is correct:** Early political associations such as the Landholders’ Society, British Indian Association, and Madras Native Association were dominated by zamindars, landlords, professionals, and educated elites.
 - ✓ Their leadership largely came from wealthy and influential sections of society.
- **Statement 2 is correct:** Most pre-Congress political organisations were confined to specific regions or presidencies.
 - ✓ They focused on local and provincial grievances rather than building an all-India political movement.
 - ✓ The emergence of the Indian National Congress in 1885 marked the beginning of an organised all-India political platform.
- **Statement 3 is incorrect:** Early associations did not demand complete independence. Their demands were moderate and constitutional, such as administrative reforms, greater Indian participation in government, and protection of Indian interests.
 - ✓ The demand for complete independence (Purna Swaraj) emerged much later.

Q 62. Ans. C

Explanation:

- The rise of the Extremist (militant nationalist) phase of the Indian freedom struggle was heavily

influenced by international events that destroyed the psychological myth of European military superiority.

- **The Russo-Japanese War (1904–1905):** Japan, a small Asian island nation, completely annihilated the formidable Russian Empire.
 - ✓ This victory had a profound, electrifying impact on the Indian psyche. It proved to Indian nationalists that an Asian country, through unity, industrialisation, and patriotism, could defeat a massive European power.
 - ✓ Coinciding perfectly with Lord Curzon’s partition of Bengal in 1905, this international victory gave Indian leaders (like Bal Gangadhar Tilak, Bipin Chandra Pal, and Lala Lajpat Rai) the immense psychological confidence needed to launch the Swadeshi and Boycott movements.

Q 63. Ans. A

Explanation:

- BIRSA-101 is an indigenous CRISPR-Cas9-based gene-editing therapy for Sickle Cell Disease, developed by Indian scientists.
- QSIP (Quantum Security Integrated Processor) is associated with India’s first quantum security chip, designed to strengthen secure communications and cybersecurity infrastructure.
- NexCAR19 is India’s first indigenous CAR-T cell therapy developed for the treatment of B-cell blood cancers..
- Bharat Gen is an AI-driven multimodal Large Language Model (LLM) initiative aimed at developing foundation models for Indian languages and contexts.

Q 64. Ans. B

Explanation:

- **Statement 1 is incorrect:** Quantum Key Distribution (QKD) does not directly encrypt or transmit messages using quantum particles. Instead, it enables the secure distribution of cryptographic keys, which are subsequently used in conventional encryption methods to secure data transmission.
- **Statement 2 is correct:** QKD relies on fundamental principles of quantum mechanics, such as the uncertainty principle and quantum entanglement, to ensure that any attempt to intercept the key can be detected.
- **Statement 3 is correct:** India demonstrated its first extensive QKD network spanning more than 500 km

under a project supported by the National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS). The achievement marked a significant step towards developing indigenous quantum-secure communication infrastructure.

Q 65. Ans. B

Explanation:

- The Economic Capital Framework (ECF) of the Reserve Bank of India provides a structured methodology for determining the appropriate level of risk buffers, contingency reserves, and surplus distribution by the RBI. It assesses various financial risks faced by the central bank, including market risk, credit risk, operational risk, and monetary policy risks, and prescribes the level of economic capital required to absorb potential losses.
- The framework, revised on the recommendations of the Bimal Jalan Committee (2019), guides the transfer of surplus profits from the RBI to the Government of India while ensuring that the central bank maintains adequate financial resilience.

Q 66. Ans. A

Explanation:

- **Statement 1 is correct:** In the parliamentary system, the council of ministers is collectively responsible to the lower house, and the loss of its confidence forces the entire ministry to resign. This collective responsibility, embodied in Article 75(3) at the Centre and Article 164(2) in the states, is the bedrock principle of the system.
- **Statement 2 is incorrect:** A separation of powers between the executive, legislature, and judiciary is the defining feature of the presidential system, not the parliamentary one. The parliamentary system is marked instead by a close fusion of the executive and the legislature.
- **Statement 3 is correct:** Ministers are drawn from and remain members of Parliament, and under Article 75(5) a minister who is not a member of either House for six consecutive months ceases to be a minister. This dual membership reflects the fusion of powers.
- **Statement 4 is incorrect:** The head of state, the President at the Centre and the Governor in the states, is only a nominal executive, while real executive authority vests in the Prime Minister or Chief Minister and the Council of Ministers. The system is also marked by majority party rule, leadership of the Prime Minister, political

homogeneity, secrecy of procedure, and the possibility of dissolution of the lower house. India adopted it because of familiarity with the model under British rule and a preference for responsible over stable government.

Q 67. Ans. A

Explanation:

- Under the absolute veto the President withholds assent and the bill never becomes law. It is normally exercised over private member bills and over government bills when the cabinet resigns before assent and the new cabinet advises against assent.
- Under the suspensive veto the President returns a bill, other than a money bill, for reconsideration; if the Houses pass it again, with or without amendments, the President must give assent. The veto is thus overridden by an ordinary majority.
- Under the pocket veto the President takes no action on a bill, as no time limit is prescribed for granting or withholding assent. President Zail Singh used it on the Indian Post Office Amendment Bill in 1986.
- The qualified veto is one that can be overridden by the legislature with a higher than ordinary majority; it is possessed by the American President and not by the Indian President. The Indian President has no veto over money bills, which can only be assented to or withheld, and after the 24th Amendment Act has no veto over constitutional amendment bills, to which assent is mandatory. Over state bills reserved by the Governor, the President enjoys an absolute veto under Articles 200 and 201.

Q 68. Ans. A

Explanation:

- Article 131 confers original jurisdiction over disputes between the Centre and states or among states, which the Supreme Court alone can decide. This jurisdiction is exclusive and does not extend to disputes arising out of pre-constitution treaties or to inter-state water disputes barred under Article 262.
- Article 143 confers advisory jurisdiction, under which the President may seek the Court's opinion on a question of law or fact of public importance, or on a pre-constitution legal matter. The opinion is not binding on the President and is not a judicial pronouncement.
- Article 137 empowers the Supreme Court to review any judgment or order made by it, which led to the evolution of the curative petition to prevent abuse of process and miscarriage of justice.

- Article 129 makes the Supreme Court a court of record with the power to punish for its contempt. Its other jurisdictions include writ jurisdiction under Article 32, appellate jurisdiction in constitutional, civil, and criminal matters under Articles 132 to 134, special leave to appeal under Article 136, and the aid of all authorities under Article 144, while Article 124 governs the appointment and removal of judges through the collegium system after the 99th Amendment establishing the National Judicial Appointments Commission was struck down in 2015.

Q 69. Ans. B

Explanation:

- The concept of public interest litigation originated in the United States in the 1960s, where it was designed to provide legal representation to unrepresented and marginalised groups. In India it was introduced in the late 1970s and early 1980s largely through Justices P. N. Bhagwati and V. R. Krishna Iyer, who relaxed the traditional rule of locus standi to allow any public-spirited person to approach the court on behalf of those unable to do so.
- The Hussainara Khatoon case of 1979, concerning undertrial prisoners in Bihar, is regarded as the first reported public interest litigation in India, while the S. P. Gupta case of 1981 formally articulated the concept. Public interest litigation can be filed under Article 32 before the Supreme Court or under Article 226 before a High Court, and the Court has even treated letters and telegrams as petitions under what is called epistolary jurisdiction. To prevent its misuse as publicity or political interest litigation, the Supreme Court has framed guidelines requiring genuine public interest and discouraging frivolous petitions.

Q 70. Ans. A

Explanation:

- **Assertion is true:** The writ jurisdiction of a High Court under Article 226 is wider in scope than that of the Supreme Court under Article 32, because Article 32 can be invoked only for the enforcement of fundamental rights.
- **Reason is true and correctly explains the Assertion:** Article 226 empowers a High Court to issue writs not only for the enforcement of fundamental rights but also for any other purpose, which means for the enforcement of ordinary legal rights as well. This

wider purpose is precisely why the High Court's jurisdiction is broader, so the Reason is the correct explanation.

- A further distinction is that the right to move the Supreme Court under Article 32 is itself a fundamental right that cannot be refused, whereas the High Court's power under Article 226 is discretionary. The Supreme Court can issue writs only against authorities within the territorial limits implied by the cause of action, and the five writs available are habeas corpus, mandamus, prohibition, certiorari, and quo warrant, each with a distinct purpose and class of respondents.

Q 71. Ans. D

Explanation:

- **Statement 1 is incorrect:** Bihar is steadily emerging as a major player in clean energy and has set a target of meeting 43.33% of its electricity requirement from renewable energy sources by 2030. To achieve this goal, the state is implementing initiatives under its **Renewable Energy Policy**, focusing on large-scale solar projects, rural mini-grids, rooftop solar installations, and the solarization of government buildings.
- **Statement 2 is incorrect:** The **Bihar Renewable Energy Development Agency (BREDA)** was established in 1987 as a registered society under the **Societies Registration Act, 1860**. Functioning under the **Department of Energy, Government of Bihar**, BREDA serves as the state's nodal agency for the promotion, planning, implementation, and monitoring of renewable energy and energy conservation programmes.

Several major solar parks are being developed across the state, including projects at **Kajra (Lakhisarai)**, **Pirpainti (Bhagalpur)**, **Jamui**, and **Banka**. The **Kajra Solar Park** is particularly significant as it is planned to host one of India's largest **Battery Energy Storage Systems (BESS)**, enabling storage of solar power and ensuring reliable electricity supply during peak demand periods.

Q 72. Ans. A

Explanation:

- Bindeshwari Dubey assumed office as the Chief Minister of Bihar on 12 March 1985 following the Congress victory in the Assembly elections and remained in office until 13 February 1988. His tenure was marked by efforts to tackle organized

crime through initiatives such as Operation Black Panther and Operation Siddhartha.

- Bhagwat Jha Azad became Chief Minister on 14 February 1988 after replacing Bindeshwari Dubey. His tenure lasted until 10 March 1989, making it one of the shorter Congress governments of the period.
- Popularly known as “Chhote Saheb”, Satyendra Narayan Sinha served as Chief Minister from 11 March 1989 to 6 December 1989. A prominent Congress leader, he had earlier served as a minister under both Binodanand Jha and K.B. Sahay.
- Lalu Prasad Yadav first became Chief Minister on 10 March 1990 after the Janata Dal's victory in Bihar. He remained Chief Minister until 25 July 1997, when he resigned following the fodder scam controversy and was succeeded by Rabri Devi.

Q 73. Ans. A

Explanation:

- **Assertion (A) is true:** The Firdausi Silsila, a branch of the Suhrawardi order, found its most prominent and successful base in medieval Bihar, making the region a major center of Sufi learning.
- **Reason (R) is true:** Makhdoom Sharafuddin Yahya Maneri (son of Makhdoom Yahya Maneri of Maner) was the greatest saint of the Firdausi order. Based in Bihar Sharif, he popularized the Silsila and spread its teachings extensively. His famous collection of letters, Maktubat-i-Sadi (The Hundred Letters), remains a highly influential text in Sufi thought. Because of his immense contribution, Bihar became the heart of the Firdausi Silsila, making (R) the correct explanation for (A).

Q 74. Ans. D

Explanation:

- Kajari is a popular monsoon folk song of the Bhojpuri region, sung during the months of Shravan and Bhadrapada. Its themes revolve around rain, love, and separation.
- Lorikayan narrates the heroic deeds of the legendary Ahir hero Lorik. It is a folk epic dominated by Veer Ras and is transmitted through oral tradition.
- Chhau is a martial dance form characterized by vigorous movements, mock combat, and battle formations, reflecting its origin in military traditions.
- Sama-Chakeva is a folk festival of the Mithila region that symbolizes affection and bonding between brothers and sisters.

Q 75. Ans. B

Explanation:

- **Assertion (A) is true:** Takht Sri Patna Sahib draws its significance from its association with the birth of Guru Gobind Singh, the tenth Sikh Guru, who was born here in 1666. This is the primary reason the shrine was established and marks it as one of the most revered seats in the Sikh tradition. The takht was later commissioned by Maharaja Ranjit Singh in the 19th century specifically to mark this birthplace.
- **Reason (R) is true:** The shrine does preserve several relics associated with Guru Gobind Singh and the early Sikh tradition, including a golden-plated cradle (pangura) used in his childhood, iron arrows, a sacred sword, a pair of sandals, and hukamnamas of Guru Gobind Singh and Guru Tegh Bahadur.
- **Why R is not the correct explanation. of A:** Assertion A locates the shrine's significance in the birth of Guru Gobind Singh, whereas Reason R points to a separate, additional source of importance, namely the preservation of relics. The relics add to the shrine's value but do not explain why the birth association makes it significant. The two statements stand as independent facts, so R is true but does not account for A.

Q 76. Ans. B

Explanation:

- The 16th Finance Commission has recommended that Bihar receive 9.95% of the divisible pool of central taxes during the award period 2026–31, making it one of the largest beneficiaries among Indian states.
 - ✓ 15th Finance Commission (2021–26): Bihar's share = 10.06%
 - ✓ 16th Finance Commission (2026–31): Bihar's share = 9.95%
 - ✓ States' overall share in divisible pool: 41% (unchanged)

Q 77. Ans. A

Explanation:

- **Statement 1 is correct:** As per the Bihar Budget 2026–27, the state's Gross State Domestic Product (GSDP) at current prices is projected to reach approximately ₹13.1 lakh crore in 2026–27. This reflects the total monetary value of all goods and services produced within Bihar during the year, measured at prevailing market prices.

- **Statement 2 is incorrect:** As per PRS, Bihar's GSDP at constant prices (which removes the effect of inflation and reflects real economic growth) is estimated to grow by 8.6% in 2024–25 over the previous year, not 6.6%. The higher growth rate indicates robust real economic expansion driven by sectors such as services, construction, and agriculture.

Q 78. 78. Ans. D

Explanation:

- **Statement 1 is correct:** Under the Education sector, Bihar has allocated ₹12,107 crore for Samagra Shiksha Abhiyan, one of the largest allocations within the education budget. The scheme aims to strengthen school education from pre-primary to Class XII through improvements in infrastructure, teacher support, digital learning, and inclusive education. Education, Sports, Arts and Culture received ₹70,141 crore in the 2026–27 Budget, making it the largest sectoral allocation.
- **Statement 2 is correct:** Under the Social Welfare and Nutrition sector, the Bihar government has provided ₹9,052 crore for Mahila Sashaktikaran (Women Empowerment). This allocation supports programmes related to women's welfare, livelihood generation, self-employment, and empowerment initiatives. The allocation follows a significantly higher revised estimate of ₹21,050 crore in 2025–26, largely due to women-centric schemes such as the Mukhyamantri Mahila Rojgar Yojana. Social Welfare and Nutrition has been allocated ₹24,710 crore in 2026–27.
- **Statement 3 is correct:** Within the Rural Development sector, Bihar has allocated ₹1,890 crore for VB-G-RAM-G, while MGNREGS has been allocated ₹3,192 crore. These allocations are aimed at strengthening rural infrastructure, livelihood generation, and employment opportunities in rural areas. The Rural Development sector received a total allocation of ₹30,387 crore in the 2026–27 Budget.

Q 79. Ans. D

Explanation:

- **Statement 1 is correct:** Bihar Budget 2026–27, Bihar is projected to record a revenue surplus of ₹1,143 crore (0.1% of GSDP) in 2026–27. This marks a significant improvement from 2025–26 (Revised Estimates), when the state is expected to register a revenue deficit of ₹76,315 crore, equivalent to 6.7%

of GSDP.

- **Statement 2 is correct:** The fiscal deficit for 2025–26 (Revised Estimates) is expected to be 11.8% of GSDP, much higher than the budget estimate. The increase is primarily attributed to a sharp rise in expenditure during the year. For 2026–27, the state has targeted a fiscal deficit of 3.0% of GSDP, in line with fiscal responsibility norms.

Key Budget Facts for Revision:

- **Revenue Surplus (2026–27):** 0.1% of GSDP
- **Revenue Deficit (2025–26 RE):** 6.7% of GSDP
- **Fiscal Deficit (2025–26 RE):** 11.8% of GSDP
- **Fiscal Deficit Target (2026–27 BE):** 3.0% of GSDP

Q 80. Ans. B

Explanation:

- **Statement 1 is incorrect:** Nalanda University was destroyed in 1193 CE during the invasion led by Muhammad bin Bakhtiyar Khilji. Bakhtiyar Khilji was a Turkic military commander who served under Qutb-ud-din Aibak and belonged to the Khilji tribe of Turkic origin. He is known for his military campaigns in eastern India, during which he conquered large parts of Bihar and Bengal.
- **Statement 2 is correct:** The site of Nalanda University was first brought to scholarly attention in 1812 by the Scottish surveyor Francis Buchanan-Hamilton. The historical significance of the site was later established in 1861 by Sir Alexander Cunningham, who correlated the archaeological remains with the accounts of Chinese pilgrims and identified the site as the famed Nalanda Mahavihara. Sir Alexander Cunningham, a British Army engineer, is widely regarded as the “Father of Indian Archaeology” and was the founder as well as the first Director-General of the Archaeological Survey of India (ASI).
- **Statement 3 is correct:** The proposal to revive the ancient Nalanda University was first put forward by former President Dr. A.P.J. Abdul Kalam in March 2006. He advocated the establishment of a modern international university near Rajgir in Bihar. The proposal received support from participating countries at the East Asia Summit in 2007 and subsequently culminated in the enactment of the Nalanda University Act, 2010, paving the way for the establishment of the new Nalanda University.
- **Statement 4 is correct:** The Archaeological Site of Nalanda Mahavihara was included in the UNESCO World Heritage List in July 2016. The remains of

the site date from the 3rd century BCE to the 13th century CE, reflecting a long tradition of learning and Buddhist monastic education. Nalanda was one of the world's earliest residential universities and attracted students and scholars from different parts of Asia. It was a renowned centre of learning where subjects such as Buddhist philosophy, the Vedas, astronomy, mathematics, logic, and medicine were taught.

Nalanda University

- Nalanda Mahavihara was founded in 427 CE by Kumaragupta I of the Gupta dynasty and was located near present-day Rajgir in Bihar. The university received patronage from several rulers, including Harshavardhana (606–648 CE) and Bhaskaravarman, the ruler of Kamarupa (Assam). Nalanda emerged as an important centre of Mahayana Buddhism and played a significant role in the spread of Buddhist learning across Asia.
- Many renowned Buddhist scholars studied and taught at Nalanda, including Nagarjuna, Dignaga, Dharmapala, Dharmakirti, Silabhadra, Shantarakshita, Padmasambhava, Atisa, Candrakirti, and Jinamitra. Among its notable scholars was Naropa, who contributed to the transmission of Buddhism to Tibet. The celebrated mathematician and astronomer Aryabhata is also believed to have served as the head of the university during the sixth century CE.

Q 81. Ans. C

Explanation:

- **Statement 1 is correct:** Certificates of Deposit (CDs) are short-term negotiable money market instruments used by banks to mobilise funds from investors. The RBI's Master Direction on Certificates of Deposit permits Scheduled Commercial Banks (SCBs) to issue CDs. These instruments are generally issued at a discount and redeemed at face value on maturity, thereby providing an additional avenue for short-term resource mobilisation by banks.
- **Statement 2 is incorrect:** Local Area Banks (LABs) are small private sector banks with a limited geographical footprint. They are not included in the RBI's master list of eligible issuers and are strictly not permitted to issue Certificates of Deposit.
- **Statement 3 is correct:** Select All-India Financial Institutions (AIFIs) that have been explicitly

permitted by the Reserve Bank of India can issue Certificates of Deposit. They use these instruments to raise short-term resources within the umbrella limits prescribed by the RBI.

- **Statement 4 is correct:** In a major regulatory update introduced in 2021, the RBI explicitly permitted Regional Rural Banks (RRBs) to issue Certificates of Deposit. This broadened their ability to mobilize short-term funds, entirely reversing the older guidelines that previously barred them from doing so.

Q 82. Ans. C

Explanation:

- **Statement 1 is correct:** The Wilmington Declaration, adopted at the QUAD Leaders' Summit held in Wilmington, Delaware (2024), announced the Maritime Initiative for Training in the Indo-Pacific (MAITRI). The initiative aims to strengthen maritime security and domain awareness by providing training and capacity-building support to partner countries in the Indo-Pacific region. It reflects the QUAD's emphasis on a free, open, inclusive, and rules-based maritime order.
- **Statement 2 is correct:** The Declaration also launched the QUAD Cancer Moonshot Initiative, a collaborative effort to reduce the burden of cancer in the Indo-Pacific. The initiative focuses on enhancing cancer prevention, screening, diagnosis, treatment, and research cooperation, particularly in low- and middle-income countries across the region.
- **Statement 3 is incorrect:** The Wilmington Declaration did not establish a permanent QUAD Secretariat. The QUAD continues to function as an informal strategic partnership among India, Australia, Japan, and the United States, without a treaty-based institutional structure or permanent secretariat. Cooperation is carried out through leaders' summits, ministerial meetings, and working groups.

Q 83. Ans. A

Explanation:

- Amrit Gyan Kosh (AGK) is a digital knowledge platform launched by the Government of India to institutionalise knowledge sharing across ministries, departments, and public institutions. It serves as a central repository of governance-related resources, including case studies, policy documents, training materials, reports, and best practices.

- The platform aims to promote evidence-based policymaking, preserve institutional knowledge, and facilitate continuous learning among civil servants and other stakeholders.

Q 84. Ans. A

Explanation:

- **Assertion (A) is true:** The **Securitisation and Reconstruction of Financial Assets and Enforcement of Security Interest (SARFAESI) Act, 2002** empowers secured creditors, such as banks and financial institutions, to enforce their security interests without first obtaining the approval of a court or tribunal. The Act was enacted to facilitate faster recovery of non-performing assets (NPAs) and to reduce delays associated with conventional judicial proceedings.
- **Reason (R) is true:** Under **Section 13** of the SARFAESI Act, when a borrower defaults and the account is classified as a Non-Performing Asset (NPA), the secured creditor may issue a demand notice requiring repayment of the outstanding dues. If the borrower fails to comply within the prescribed period, the creditor can take possession of the secured assets, assume management of the business in certain cases, or sell the secured assets to recover the dues, following the procedure laid down in the Act.
- **Reason (R) correctly explains Assertion (A):** The authority granted under **Section 13** to take possession and realise secured assets without prior intervention of courts is precisely what enables secured creditors to enforce security interests without prior judicial approval. Thus, the mechanism provided under **Section 13** forms the legal basis of the assertion.

Q 85. Ans. D

Explanation:

- **Statement 1 is correct:** **Category I AIFs** invest in sectors or areas that are considered socially or economically desirable and are expected to have a positive spillover effect on the economy. These include venture capital funds, SME funds, social impact funds, infrastructure funds, and angel funds. Since such investments promote entrepreneurship, innovation, infrastructure development, and social welfare, regulators may provide incentives to encourage these funds.
- **Statement 2 is correct:** **Category II AIFs** comprise funds that do not fall under **Category I** or **Category**

III and generally include private equity funds, debt funds, and fund of funds. These funds are not permitted to undertake leverage or borrowing for investment purposes. However, they may borrow temporarily to meet day-to-day operational requirements, subject to limits prescribed by SEBI regarding duration, frequency, and amount.

- **Statement 3 is correct:** **Category III AIFs** employ diverse and complex trading strategies with the objective of generating short-term returns. These funds, such as hedge funds, are permitted to use leverage, including through investments in derivatives and other sophisticated financial instruments. The use of leverage enables them to amplify returns, although it also increases investment risk.

Q 86. Ans. A

Explanation:

- **Statement 1 is correct:** **Mitochondrial Replacement Therapy (MRT)** is an advanced assisted reproductive technology developed to prevent the transmission of inherited mitochondrial disorders from mother to child. These disorders arise due to mutations in mitochondrial DNA (mtDNA), which can affect energy production in cells and lead to severe neurological, muscular, or metabolic diseases.
- **Statement 2 is incorrect:** In **Pronuclear Transfer (PNT)**, fertilisation occurs first in both the mother's egg and the donor's egg. Subsequently, the pronuclei containing the nuclear genetic material from the biological parents are transferred into the donor embryo containing healthy mitochondria. Therefore, mitochondrial replacement in PNT takes place after fertilisation, not before it. The technique performed before fertilisation is known as **Maternal Spindle Transfer (MST)**.
- **Statement 3 is correct:** A child born through MRT may inherit genetic material from three individuals. The vast majority of genetic material (more than 99%) comes from the biological mother and father through nuclear DNA, while a small amount of mitochondrial DNA is inherited from the female donor who provides healthy mitochondria.
- **Statement 4 is incorrect:** Mitochondrial DNA is transmitted almost exclusively through the maternal lineage, because the mitochondria present in the sperm generally do not contribute to the embryo after fertilisation. Consequently, mitochondrial diseases are typically inherited from the mother rather than the father.

Q 87. Ans. A

Explanation:

- **Statement 1 is correct:** Jevons Paradox refers to the phenomenon whereby improvements in the efficiency with which a resource is used can lead to an increase, rather than a decrease, in the overall consumption of that resource. First observed by economist William Stanley Jevons in the context of coal use, the paradox highlights how lower costs and greater efficiency may stimulate higher demand.
- **Statement 2 is incorrect:** The Solow Paradox, associated with economist Robert Solow, states the opposite of what is given in the pair. It refers to the observation that substantial advances in information technology were not immediately reflected in productivity statistics. The famous remark, "You can see the computer age everywhere but in the productivity statistics," captures this paradox. Therefore, technological progress does not always result in proportional and immediate productivity gains.
- **Statement 3 is correct:** AlphaFold, developed by DeepMind, is an artificial intelligence system that predicts the three-dimensional structure of proteins from their amino acid sequences. The breakthrough has significantly accelerated research in molecular biology, drug discovery, and biomedical sciences.
- **Statement 4 is correct:** C2S-SCALE (Cell-to-System Scale) is an AI-driven biological modelling initiative designed to understand and interpret interactions occurring within and between cells. It aims to integrate biological data across multiple scales to improve understanding of complex biological systems and disease mechanisms.

Q 88. Ans. D

Explanation:

- **Statement 1 is correct:** A rectangular drainage pattern primarily develops on a strongly jointed, fractured, or faulted rocky terrain. The underlying bedrock structure dictates the flow of water, as rivers and streams naturally seek the paths of least resistance, which in this case are the faults and joints in the rocks.
- **Statement 2 is correct:** Because the geological faults and joints typically intersect each other at right angles (like a grid), the main streams and their tributaries are forced to follow these geometric structural depressions. This causes the streams to make sharp, right-angle bends, creating a rectangular network.

- **Statement 3 is correct:** The Vindhyan Range in central India is heavily composed of jointed and faulted sedimentary rocks (such as sandstones and limestones). Because of this grid-like bedrock structure, rivers flowing through this region (like the Betwa and Ken) exhibit a classic rectangular drainage pattern.
- While both trellis and rectangular patterns involve right-angle junctions, a trellis pattern develops in areas of alternating hard and soft rock layers (folded topography), whereas a rectangular pattern develops in homogeneous but heavily jointed/faulted rock.

Q 89. Ans. D

Explanation:

- Under Sir Edmund Halley's classical thermal concept of the monsoon (which characterizes the monsoon as a massive, seasonal sea breeze driven by the differential heating of land and water), the heavily moisture-laden winds blow from the ocean toward the continental landmass. When this warm, moist air current encounters structural barriers like mountains or plateaus (such as the Western Ghats or the Himalayas), it undergoes a strict sequence of thermodynamic processes known as orographic lifting:
 - ✓ **Ascent:** The physical barrier forces the advancing moist air mass to rise along its windward slope.
 - ✓ **Saturation:** As the air rises, it expands and cools adiabatically (because atmospheric pressure decreases with altitude). When the air cools down to its dew point temperature, its relative humidity reaches 100%, meaning it is fully saturated.
 - ✓ **Condensation:** Further forced ascent and cooling cause the water vapor in the saturated air to condense into microscopic liquid water droplets, leading to the formation of dense clouds.
 - ✓ **Precipitation:** Once these droplets coalesce and become too heavy for the updrafts to support, they fall as rain (orographic precipitation).

Q 90. Ans. C

Explanation:

- **Statement 1 is correct:** In a normal year, the cold Humboldt (Peru) Current keeps the western coast of South America highly arid. However, during an El Niño event, the easterly trade winds weaken,

allowing the warm equatorial waters of the western Pacific to slosh eastward toward South America. This anomalous warming of the sea surface off the coast of Peru and Chile leads to intense thermal convection, cloud formation, and unusually heavy precipitation, often resulting in severe flooding in these traditionally dry regions.

- **Statement 2 is correct:** As the massive pool of warm water and its associated atmospheric convection (rainfall) shift toward the central and eastern Pacific, the western Pacific is left depleted. The oceanic waters around Eastern Australia and Indonesia become cooler than average, creating a strong high-pressure zone in the atmosphere. This high-pressure system causes the air to sink (subsidence), which heavily suppresses cloud formation and leads to severe drought conditions, crop failures, and devastating bushfires in Australia and Indonesia.

Q 91. Ans. D

Explanation:

- **Statement 1 is correct:** Tropical cyclones develop only over large bodies of warm ocean water. They cannot form over land masses. Once a tropical cyclone crosses onto land (a process called landfall), it gets cut off from its primary moisture and heat source and quickly dissipates due to increased surface friction and lack of moisture.
- **Statement 2 is correct:** The primary thermodynamic engine of a tropical cyclone is the **latent heat of condensation**. When water evaporates from the warm ocean surface, it absorbs heat. As this warm, moist air rises and expands, it cools, causing the water vapor to condense into heavy rain clouds. This condensation releases massive amounts of latent heat into the atmosphere, which further warms the air, causing it to rise faster, lowering surface pressure, and drawing in even more moisture in a self-sustaining loop.
- **Statement 3 is correct:** A continuous and abundant supply of warm, moist air is the basic fuel for a cyclone. This is why they only form when the sea surface temperature (SST) is exceptionally warm, typically **27°C (80°F) or higher**, to a depth of at least 50 meters.
- **Statement 4 is correct:** An unstable vertical temperature profile in the troposphere is absolutely crucial. "Instability" means that the atmospheric temperature drops rapidly with height. This allows the warm, moist parcel of air from the ocean

surface to continue rising rapidly without being suppressed, leading to the formation of towering cumulonimbus clouds.

Q 92. Ans. C

Explanation:

- The passage describes the Sub-Humid Continental climate, which is the defining climatic characteristic of the Upper Gangetic Plain (Agro-Climatic Zone V, as classified by the Planning Commission of India).
 - ✓ **Geographical Area:** This zone primarily encompasses the central and western parts of Uttar Pradesh (the Ganga-Yamuna Doab and Rohilkhand regions) and parts of Uttarakhand.
 - ✓ **Temperature (Continental Effect):** Because the region is entirely landlocked and situated far from the moderating influence of the ocean, it exhibits a classic "continental" temperature profile with high seasonal extremes. Summers are intensely hot (averaging 25°C to 40°C, and occasionally higher), while winters are noticeably cooler.
 - ✓ **Rainfall:** It receives moderate average annual rainfall between 75 cm and 150 cm, almost entirely brought by the Bay of Bengal branch of the Southwest Monsoon. The rainfall volume distinctly decreases as the monsoon winds travel from east to west across this plain.
- **Sub-Humid Littoral (Maritime):** "Littoral" refers to coastal areas. This climate type describes regions with high humidity and highly moderated, equable temperatures throughout the year due to the sea's influence (e.g., the Eastern Coastal Plains/Coromandel Coast).
- **Semi-Arid Subtropical:** This climate applies to the regions further west, specifically the Trans-Ganga Plains (Punjab, Haryana, and eastern Rajasthan), where the average annual rainfall drops significantly below 75 cm and vegetation transitions to scrub and thorny bushes.
- **Sub-Humid Transition:** While the Gangetic plain acts broadly as a transition between the wet east and dry west, "Sub-Humid Continental" is the strict, formal geographic and agro-climatic terminology used by geographers to define the Upper Ganga Plain.

Q 93. Ans. A

Explanation:

- The majority of India's forest cover consists of broad-leaved species, which are climatically

adapted to the tropical and subtropical monsoon conditions of the subcontinent.

- ✓ **Statistical Dominance:** According to the Forest Survey of India (FSI), broad-leaved forests account for approximately 94% of the country's total forest area. The remaining ~6% consists of coniferous (needle-leaf) forests, which are strictly confined to the higher altitudes of the Himalayas.
- ✓ **Climatic Nature:** These broad-leaved forests are predominantly Tropical Deciduous Forests (widely known as Monsoon Forests). They thrive in regions receiving rainfall between 70 cm and 200 cm. Because of the distinct dry season in the Indian monsoon cycle, these broad-leaved trees shed their leaves for about six to eight weeks in early summer to conserve water.
- ✓ **Key Species:** This category includes some of India's most economically and ecologically significant trees, such as Teak (*Tectona grandis*), Sal (*Shorea robusta*), Shisham, Mahua, Peepal, and Neem.

Q 94. Ans. B

Explanation:

- **Statement 1 is correct:** Hydraulic action is the purely mechanical process of erosion caused by the sheer force and kinetic energy of moving water. As river water flows rapidly, it crashes against the river banks and bed, sweeping away loose material. It also forces water and air into cracks and joints in the bedrock; the compressed air exerts immense pressure, progressively weakening and breaking the rocks apart.
- **Statement 2 is incorrect:** Hydraulic action does **not** require a sediment load to cause erosion; it relies entirely on the pressure and drag of the water itself. The process described in Statement 2 is actually **Abrasion (or Corrasion)**. Abrasion is the "sandpaper effect" where the river uses its sediment load (pebbles, sand, and boulders) as grinding tools to scrape, scour, and erode the riverbed and banks.

Q 95. Ans. C

Explanation:

- **Statement 1 is correct:** In 2016, Ameenpur Lake became the very first water body (and urban lake) in India to be officially declared a Biodiversity Heritage Site (BHS) under Section 37 of the Biological Diversity Act, 2002. (Note: The first overall BHS in India was the Nallur Tamarind Grove in Karnataka, declared in 2007).

- **Statement 2 is correct:** Ameenpur Lake is located in the Sangareddy district, which is situated in the western part of Telangana (on the north-western outskirts of the Hyderabad metropolitan area).
- It is a 300-year-old, man-made reservoir constructed during the reign of Ibrahim Quli Qutb Shah (16th century) to irrigate the surrounding gardens.
- Despite facing immense urban pressure, the lake acts as a crucial wintering ground and habitat for over 150 species of resident and migratory birds, including greater flamingos, spot-billed pelicans, painted storks, and kingfishers.

Q 96. Ans. B

Explanation:

- **Statement 1 is correct:** Estuaries are ecotones (transition zones) between freshwater riverine ecosystems and saline marine ecosystems. Because they trap a massive continuous supply of nutrients brought down by rivers, combined with the churning action of ocean tides, they are incredibly fertile. They rank among the most biologically productive ecosystems on Earth, often surpassing even tropical rainforests in terms of biomass production.
- **Statement 2 is incorrect:** Estuaries are exclusively formed at the lower course (the mouth) of rivers, not the upper course. The upper course is the river's mountainous source where it originates. An estuary is strictly a coastal feature where the river flows directly into the sea or ocean.
- **Statement 3 is correct:** By definition, an estuary is a semi-enclosed coastal body of water with a free connection to the open sea. Because of this connection, it is continuously subjected to tidal fluctuations, which causes the daily mixing of lighter freshwater from the river and denser saltwater from the ocean (creating brackish water).
- The rivers flowing into the Bay of Bengal (like the Ganga, Godavari, and Krishna) predominantly form deltas due to their heavy sediment load and the gentle coastal gradient. Conversely, the west-flowing rivers into the Arabian Sea (like the Narmada and Tapi) form estuaries because they flow through hard rift valleys, carry less sediment, and meet the sea at a steeper gradient.

Q 97. Ans. B

Explanation:

- The Labrador Basin is a major oceanic basin located in the extreme northwestern region of the North Atlantic Ocean.

- **Geographical Location:** It is strategically situated between the continental shelf of Greenland to the northeast and the continental margins of the Labrador Peninsula and Newfoundland (Canada) to the southwest.
- **Formation:** The basin was formed by the process of seafloor spreading that occurred when the North American Plate and the Greenland Plate rifted apart millions of years ago (during the Late Cretaceous to Early Cenozoic eras).
- The Labrador Basin (and the Labrador Sea) plays a critical global role in the Earth's climate system. It is one of the few places in the world where deep-water formation occurs. In winter, frigid winds cool the surface water, increasing its density and causing it to sink. This sinking water forms the Labrador Sea Water, a vital driving component of the Atlantic Meridional Overturning Circulation (AMOC).

Q 98. Ans. C

Explanation:

- **Pair 1 is correctly matched:** The platypus (along with the echidna) is a unique type of mammal that lays eggs instead of giving birth to live young. These are known as monotremes. Despite laying eggs, they possess mammary glands to nourish their offspring, which firmly classifies them as mammals.
- **Pair 2 is correctly matched:** Amphibians (such as frogs, toads, and salamanders) possess a **three-chambered heart**, consisting of two atria and one shared ventricle. This structure results in some mixing of oxygenated and deoxygenated blood.
- **Pair 3 is incorrectly matched:** Fishes are **cold-blooded** (poikilothermic) animals. Their internal body temperature is not constant; instead, it dynamically fluctuates to match the temperature of the surrounding water. Furthermore, fishes only have a two-chambered heart.
- **Pair 4 is correctly matched:** Birds (Aves) have a fully divided **four-chambered heart** (two atria and two ventricles). This prevents any mixing of oxygen-rich and oxygen-poor blood, providing the highly efficient oxygen delivery system required to support their warm-blooded (homeothermic) metabolism and the immense physical demands of flight.

Q 99. Ans. B

Explanation:

- In the **saprotrophic** (or saprophytic) mode of nutrition, organisms obtain their nutrients from dead and decaying organic matter. They do this by

secreting digestive enzymes outside their bodies to break down complex organic materials into simpler soluble substances, which they then absorb. Common examples include fungi (like mushrooms, bread mould, and yeast) and many soil bacteria.

- **Parasitic:** The organism (parasite) lives inside or on the outside of another living organism (the host) and derives its nutrition directly from it, often harming the host in the process. Examples include ticks, lice, tapeworms, and the plant *Cuscuta* (Amarbel).
- **Autotrophic:** The organism synthesizes its own food from simple inorganic substances (like carbon dioxide and water) using external energy, typically sunlight. Green plants and certain bacteria (cyanobacteria) are autotrophs.
- **Holozoic:** The organism ingests solid, complex organic food, which is then broken down and digested internally. Humans, dogs, and even unicellular organisms like *Amoeba* utilize holozoic nutrition.

Q 100. Ans. B

Explanation:

- A sharp knife cuts more effectively because it exerts a much greater pressure on the object due to its significantly smaller surface area.
- Pressure is defined as the physical force exerted on an object per unit area. The mathematical relationship is $\text{Pressure} = \text{Force} / \text{Area}$ (or $P = F/A$).
- **Sharp vs. Blunt:** A sharp knife is ground to an extremely thin edge, meaning the area of contact with the object is microscopic. Because area is the divisor in the equation, decreasing the area produces a vastly higher pressure for the exact same amount of applied force. This highly concentrated pressure easily pierces and yields the material. Conversely, a blunt knife has a thicker, rounded edge (larger area), which spreads the same force over a larger surface, resulting in lower pressure that struggles to cut.
- **Pointed Nails/Pins:** Iron nails and drawing pins have sharp, pointed tips so that the force exerted by a hammer (or thumb) produces enough concentrated pressure to easily pierce wood or walls.
- **Wide Bag Straps:** School bags and heavy backpacks are designed with wide straps to increase the area of contact on the shoulders, which reduces the overall pressure and makes them more comfortable to carry.
- **Camel's Feet:** Camels have broad, flat feet. This increases the surface area in contact with the ground, reducing the downward pressure and preventing them from sinking into soft desert sand.

Q 101. Ans. B**Explanation:**

- 2020 is a leap year, so it has 366 days.
- $366 = 52 \text{ weeks} + 2 \text{ odd days}$.
- Wednesday + 2 = Friday.
- Hence 1 January 2021 was a Friday.

Q 102. Ans. A**Explanation:**

Logic: Each letter is shifted forward by +3 positions in the alphabet.

$B+3=E$, $I+3=L$, $R+3=U$, $D+3=G$

Hence BIRD = ELUG.

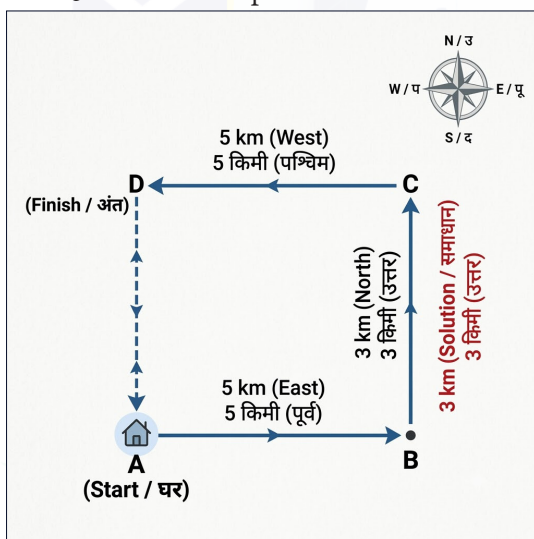
Q 103. Ans. A**Explanation:**

Starting Point: Let Ravi's house be at point A.

First Move: He walks 5 km towards the East to reach point B.

Second Move: He turns left (which faces North) and walks 3 km to reach point C.

Third Move: He turns left again (which faces West) and walks 5 km to reach point D.



Since $AB = 5\text{km}$ and $CD = 5\text{ km}$, points A and D lie on the same vertical line.

The distance between his starting point (A) and final point (D) is equal to the distance BC.

Distance = $AD = BC = 3\text{ km}$

Looking from the starting point A, point D is directly above it.

Answer: Ravi is 3 km North from his starting point.

Q 104. Ans. A**Explanation:**

Alphabetical order: Space, Spare, Speak, Speed, Splash.

5 (Space) → 2 (Spare) → 3 (Speak) → 4 (Speed) → 1 (Splash).

Correct order: 5, 2, 3, 4, 1.

Q 105. Ans. C**Explanation:**

Pattern: Each cluster has letters with a gap of +2.

BDFH: 2,4,6,8 — gap of 2 ✓

MOQS: 13,15,17,19 — gap of 2 ✓

PRTU: 16,18,20,21 — last gap is 1, not 2 ✗

TVXZ: 20,22,24,26 — gap of 2 ✓

Hence PRTU is the odd one out.

Q 106. Ans. C**Explanation:**

Let principal = P

Amount = $P + (P \times 5 \times 4)/100 = P(1 + 0.20) = 1.2P$

$1.2P = 1320$

$P = 1320 / 1.2 = \text{Rs. } 1100$

Hence the correct answer is Rs. 1100.

Q 107. Ans. C**Explanation:**

SP = Rs. 480, Loss = 20%

CP = $480 / 0.8 = \text{Rs. } 600$

To gain 20%: SP = $600 \times 1.2 = \text{Rs. } 720$

Hence the correct answer is Rs. 720.

Q 108. Ans. C**Explanation:**

$12\text{ CP} = 9\text{ SP}$

CP/SP = $9/12 = 3/4$

Profit % = $(\text{SP} - \text{CP})/\text{CP} \times 100 = (4 - 3)/3 \times 100 = 33.33\%$

Hence the correct answer is 33.33%.

Q 109. Ans. B**Explanation:**

SI = Amount - Principal = $2600 - 2000 = \text{Rs. } 600$

SI = $(P \times R \times T) / 100$

$600 = (2000 \times R \times 3) / 100$

$R = 60000 / 6000 = 10\%$

Hence the correct answer is 10%.

Q 110. Ans. B**Explanation:**

$A = P(1 + R/100)^T$

$A = 5000 \times (1 + 10/100)^2 = 5000 \times 1.1 \times 1.1 = 5000 \times 1.21 = \text{Rs. } 6050$

CI = $6050 - 5000 = \text{Rs. } 1050$

Hence the correct answer is Rs. 1050.

Q 111. Ans. A

Explanation:

- **Statement 1 is correct:** Lok Adalats were given statutory status by the Legal Services Authorities Act of 1987, which institutionalised this mechanism of alternative dispute resolution and also established the National Legal Services Authority to provide free legal aid and organise such forums.
- **Statement 2 is incorrect:** An award of a Lok Adalat is deemed to be a decree of a civil court, is final and binding on the parties, and no appeal lies against it before any court. There is also no court fee, and any fee already paid is refunded if the dispute is settled.
- Lok Adalats handle pending cases and pre-litigation disputes that are compoundable or capable of compromise, but cannot decide non-compoundable criminal cases. The Permanent Lok Adalats, established under a 2002 amendment for public utility services such as transport, postal, and telegraph services, have adjudicatory powers and can decide a dispute on merits even if the parties fail to reach a settlement, up to a prescribed pecuniary limit. Related forums include Family Courts under the 1984 Act and Gram Nyayalayas under the 2008 Act.

Q 112. Ans. A

Explanation:

- **Statement 1 is correct:** The Goods and Services Tax Council is a constitutional body established under Article 279A, which was inserted by the 101st Constitutional Amendment Act of 2016 to administer the goods and services tax through a joint forum of the Centre and the states.
- **Statement 2 is correct:** The Union Finance Minister is the ex officio Chairperson of the Council, and its members include the Union Minister of State in charge of Finance or Revenue and the minister in charge of finance or taxation of each state, with a vice chairperson chosen from among the state ministers.
- **Statement 3 is incorrect:** A decision of the Council is taken by a majority of not less than three fourths of the weighted votes of the members present and voting, not two thirds. In this weighting the vote of the Centre carries one third of the total votes cast, while the votes of all the states together carry two thirds, and the quorum is one half of the total members. The functions of the Council cover tax rates, exemptions, threshold limits, model laws,

and special provisions, and the Supreme Court in the Mohit Minerals case of 2022 held that the recommendations of the Council are persuasive and not binding on the Union and the states.

Q 113. Ans. C

Explanation:

- **Statement 1 is correct:** The National Investigation Agency was constituted under the National Investigation Agency Act of 2008, enacted in the aftermath of the Mumbai terror attacks of November 2008, to act as the central agency for the investigation and prosecution of offences affecting the sovereignty, security, and integrity of the country.
- **Statement 2 is correct:** The agency can investigate the scheduled offences listed in the Act, such as those under the Unlawful Activities Prevention Act and the Atomic Energy Act, without the prior consent of the state government, unlike the Central Bureau of Investigation, which requires state consent to operate in a state. The Central Government can direct the agency to take up a case, and its officers enjoy powers equivalent to those of state police officers.
- The cases of the agency are tried by special courts designated for the purpose, and appeals lie to the High Court. The National Investigation Agency Amendment Act of 2019 widened its scope by allowing it to investigate scheduled offences committed outside India, subject to international law, and added offences such as human trafficking, counterfeit currency, manufacture or sale of prohibited arms, cyber terrorism, and offences under the Explosive Substances Act to its jurisdiction.

Q 114. Ans. A

Explanation:

- **Statement 1 is correct:** The doctrine of pleasure, under which a civil servant holds office during the pleasure of the President at the Centre or the Governor in the states under Article 310, is borrowed from the British common law, where a servant of the Crown holds office at the pleasure of the Crown.
- **Statement 2 is correct:** The doctrine is qualified by Article 311, which provides that a civil servant cannot be dismissed or removed by an authority subordinate to the one that appointed him, and cannot be dismissed, removed, or reduced in rank

except after an inquiry giving him a reasonable opportunity of being heard, subject to three exceptions involving conviction on a criminal charge, impracticability of holding an inquiry, and the satisfaction of the President or Governor that an inquiry is not expedient in the interest of the security of the state.

- **Statement 3 is incorrect:** The doctrine of pleasure does not apply to judges of the Supreme Court and High Courts, the Chief Election Commissioner, the Comptroller and Auditor General, and the Chairman and members of the Union Public Service Commission, all of whom enjoy security of tenure and can be removed only through the special procedures prescribed for them. Services in India are classified into All India Services, Central Services, and State Services, and the All India Services, such as the Indian Administrative Service and the Indian Police Service, are governed by Article 312, under which Parliament can create a new All India Service only after the Rajya Sabha passes a resolution supported by two-thirds of the members present and voting.

Q 115. Ans. A

Explanation:

- The Gujral Doctrine, enunciated by I. K. Gujral in 1996, set out five principles of conduct with India's neighbours, the central one being non-reciprocity, under which India would extend accommodation to smaller neighbours such as Bangladesh, Bhutan, the Maldives, Nepal, and Sri Lanka without expecting an equal return.
- Panchsheel refers to the five principles of peaceful coexistence first formally stated in the 1954 agreement between India and China, namely mutual respect for territorial integrity and sovereignty, mutual non-aggression, mutual non-interference in internal affairs, equality and mutual benefit, and peaceful coexistence.
- The Connect Central Asia Policy, articulated in 2012, aimed at deepening political, economic, security, and cultural engagement with the Central Asian republics through a broad-based approach.
- The Act East Policy, announced in 2014, was a reorientation and intensification of the Look East Policy that had been launched in 1992 under P. V. Narasimha Rao, extending India's engagement from South East Asia to the wider Asia Pacific. The broader principles of Indian foreign policy include non-alignment, anti-colonialism, the promotion

of international peace and security under Article 51, and a no-first-use posture under the nuclear doctrine adopted in 2003.

Q 116. Ans. A

Explanation:

- The Faraizi Movement was an important Islamic reform and revivalist movement that emerged in Eastern Bengal (present-day Bangladesh) during the early nineteenth century.
- Haji Shariatullah founded the Faraizi Movement in 1818 after returning from Arabia, where he had spent several years studying Islamic theology and religious practices.
 - ✓ The term "Faraizi" is derived from the Arabic word Faraiz, meaning the obligatory duties prescribed by Islam.
 - ✓ Shariatullah emphasised strict observance of the fundamental obligations of Islam, such as prayer, fasting, and other religious duties.
 - ✓ He opposed practices that he regarded as un-Islamic innovations, superstitions, and customs that had entered local religious life.
 - ✓ The movement gained considerable support among the Muslim peasantry of Eastern Bengal.
- It also developed a social dimension by mobilising peasants against oppressive landlords and exploitative revenue practices.

Q 117. Ans. A

Explanation:

- **Statement 1 is correct:** Traditional education largely neglected the study of physical and natural sciences.
 - ✓ The curriculum of both Hindu and Muslim institutions was heavily oriented towards religious, literary, and philosophical subjects.
 - ✓ Hindu centres of learning emphasised the Vedas, Vedangas, Smritis, Nyaya, Mimamsa, Grammar, and Philosophy.
 - ✓ Muslim institutions focused on the Quran, Hadith, Islamic jurisprudence, Persian literature, and theology.
 - ✓ Scientific subjects such as modern physics, chemistry, biology, and experimental sciences were generally absent from the curriculum.
- **Statement 2 is correct:** Hindu higher education was based on Sanskrit and was largely confined to Brahmins.
 - ✓ Higher Hindu learning was imparted mainly through tools and chatuspathis.

- ✓ The medium of instruction was predominantly Sanskrit.
- ✓ Subjects included grammar, logic, law, philosophy, and religious texts.
- ✓ Access to advanced Sanskrit education was generally restricted to the Brahmin community, although exceptions existed in some regions.
- ✓ As a result, higher education remained limited to a relatively small section of society.
- **Statement 3 is incorrect:** Elementary Muslim education was usually imparted through maktabas, often attached to mosques or maintained by local communities and charitable endowments.
 - ✓ Teachers were generally appointed by local patrons, religious authorities, or community institutions.
 - ✓ Although Qazis (judicial officers) were important figures in Islamic society, they did not exercise direct and exclusive administrative control over elementary education.
 - ✓ The educational system functioned largely through local religious and community support rather than through a centralised state-controlled mechanism.
- The traditional system of education in 18th-century India was primarily organised through pathshalas, tols, chatuspathis, maktabas, and madrasas. While it preserved important literary and religious traditions, it also reflected the social and intellectual priorities of the period.

Q 118. Ans. C

Explanation:

- The Battle of Buxar (1764) was one of the most decisive battles in Indian history. It marked a major turning point in the expansion of the British East India Company and laid the foundation of British political supremacy in eastern and northern India.
- **Statement 1 is correct:** By the time the Battle of Buxar took place on October 22, 1764, Mir Qasim was no longer the Nawab of Bengal. Due to conflicts over internal trade duties and the misuse of Dastaks (free passes) by Company servants, war had already broken out between Mir Qasim and the British in 1763. Mir Qasim was defeated in a series of battles (Katwa, Murshidabad, Giria, Sooty, and Munger) and forced to flee to Awadh. In July 1763, the British reinstated his father-in-law, Mir Jafar, as the Nawab. Therefore, at Buxar, Mir Qasim fought as a deposed ruler trying to reclaim his throne.
- **Statement 2 is correct:** The allied Indian forces

consisted of the deposed Mir Qasim, the Nawab of Awadh (Shuja-ud-Daula), and the titular Mughal Emperor (Shah Alam II). Following their crushing defeat by Major Hector Munro, their fates diverged completely under the Treaties of Allahabad (August 1765):

- ✓ Mir Qasim fled the battlefield, lost all his wealth, and died in obscurity and extreme poverty near Delhi years later.
- ✓ Shah Alam II became a British pensioner. He had no real territorial power prior to the battle and was subsequently confined to Allahabad under British “protection” in exchange for granting the Diwani of Bengal, Bihar, and Orissa to the Company.
- ✓ Shuja-ud-Daula was the only one to retain his territorial domain. Robert Clive realised that annexing Awadh would expose the Company’s borders to Maratha and Afghan attacks. Thus, Awadh was restored to Shuja-ud-Daula (minus the districts of Kora and Allahabad, which were given to the Emperor), functioning as a crucial buffer state. He was also forced to pay an indemnity of 50 lakh rupees and accept a British Resident.

Q 119. Ans. A

Explanation:

- **Assertion (A) is correct:** In 1639, Francis Day, an agent of the English East India Company, acquired the lease of a coastal village called Madrasapatnam. The English quickly and relatively easily constructed a fortified factory there, which was named Fort St. George (completed around 1640). This became the very first fortified English settlement in India and the headquarters for their Coromandel Coast trade.
- **Reason (R) is correct and perfectly explains (A):** The ease with which the English fortified Madras was a direct result of the geopolitical vacuum in South India at the time. Following the catastrophic Battle of Talikota (1565), the once-mighty Vijayanagara Empire had effectively collapsed and fragmented into numerous weak, warring regional principalities ruled by local Nayaks.
 - ✓ In northern and western India, the formidable Mughal Empire strictly prohibited European merchants from building military fortifications around their trading posts (as seen in their early struggles at Surat and Hooghly).
 - ✓ The deep south lacked any such strong, centralized Indian government. The local

ruler, Damarla Venkatadri Nayaka (the Nayak of Kalahasti and a feudatory of the Chandragiri ruler), actually encouraged the English to settle and fortify the area. He wanted to boost local trade, secure customs revenues, and gain potential leverage against his rival Nayaks.

Q 120. Ans. B

Explanation:

- Before 1757, the British East India Company functioned primarily as a commercial organisation whose main objective was to procure valuable Indian goods for export to European markets. During this period, the Company had not yet acquired political control over Indian territories and depended on trade for its profits.

✓ Textiles:

- Indian cotton and silk textiles enjoyed immense demand in Europe because of their superior quality, durability, and craftsmanship.
- Bengal was famous for its fine muslin and silk fabrics.
- The Coromandel Coast produced high-quality painted and printed cotton textiles known as chintz and calicoes.
- Indian textiles constituted the largest share of the Company's exports during the seventeenth and early eighteenth centuries.

✓ Spices:

- Pepper, cardamom, ginger, and other spices were highly valued in European markets.
- Although the Dutch dominated much of the Southeast Asian spice trade, Indian spices remained an important component of Company commerce.

Q 121. Ans. B

Explanation:

- **Statement 1 is correct:** The Phenome India-CSIR Health Cohort Knowledgebase (PI-CHeCK) is a nationwide longitudinal health study launched by CSIR to create a comprehensive database linking genetic, lifestyle, environmental, and clinical factors. It is designed to track the health trajectories of individuals over time and generate insights into disease patterns, risk factors, and preventive healthcare strategies specific to the Indian population.

- **Statement 2 is incorrect:** The proposed National Biobank draws inspiration from international biobanking initiatives such as the UK Biobank, but it is not a mere replica of that model. It is being developed with modifications tailored to India's unique genetic diversity, disease burden, demographic characteristics, environmental conditions, and public health requirements. The objective is to create a repository that reflects the specific biomedical and epidemiological realities of India.

- **Statement 3 is correct:** The National Biofoundry Network is one of the key initiatives envisaged under the BioE3 Policy (Biotechnology for Economy, Environment and Employment). The network aims to establish advanced facilities for synthetic biology, biomanufacturing, and biotechnology innovation, thereby accelerating the development of bio-based products and supporting India's transition towards a high-performance bioeconomy.

Q 122. Ans. C

Explanation:

- **Veerangana Durgavati Tiger Reserve** is located in **Madhya Pradesh**. It was notified as the state's **eighth tiger reserve** and has been named after **Rani Durgavati**, the celebrated Gond queen who is remembered for her resistance against Mughal expansion in central India.
- The tiger reserve has been created by combining the **Nauradehi Wildlife Sanctuary** and the **Durgavati Wildlife Sanctuary** in the Sagar, Damoh, Narsinghpur, and Raisen districts of Madhya Pradesh. Spread across a large landscape in the central Indian highlands, the reserve is expected to strengthen tiger conservation by improving habitat connectivity between existing tiger reserves such as **Panna, Satpura**, and other protected areas in the region.

Q 123. Ans. A

Explanation:

- **Statement 1 is correct:** India became a Sectoral Dialogue Partner of ASEAN in 1992 and was elevated to the status of a Full Dialogue Partner in 1995. Subsequently, India became a Summit-level Partner in 2002 when it participated in the first ASEAN-India Summit. Thus, India attained Dialogue Partnership status before becoming a Summit-level Partner.
- **Statement 2 is correct:** Timor-Leste officially

concluded its accession process and was formally admitted as the 11th full member state of ASEAN during the 47th ASEAN Summit in Kuala Lumpur in late 2025.

- **Statement 3 is incorrect:** The ASEAN Secretariat is located in Jakarta, Indonesia, and serves as the administrative and coordinating centre of the organisation. It supports the implementation of ASEAN policies, programmes, and agreements among member states. Kuala Lumpur is not the headquarters of the Secretariat.

Q 124. Ans. B

Explanation:

- Off-budget borrowing refers to borrowings undertaken by government-owned entities such as public sector undertakings (PSUs), special purpose vehicles, or other public agencies on behalf of the government to finance its expenditure, while the associated liabilities are not directly reflected in the Union Government's budget or fiscal deficit calculations.
- Governments often use this mechanism to fund expenditures such as food, fertiliser, infrastructure, or welfare schemes without immediately increasing the reported fiscal deficit. For instance, a public sector entity may borrow from the market and utilise the funds for activities that would otherwise have required direct budgetary support from the government. Although such borrowings may not appear in the fiscal deficit in the year they are incurred, they ultimately create liabilities that are borne by the government either explicitly or implicitly.
- The N.K. Singh Committee on FRBM Review and the Comptroller and Auditor General (CAG) have highlighted that extensive reliance on off-budget borrowings can reduce fiscal transparency by understating the government's actual debt and borrowing requirements.

Q 125. Ans. B

Explanation:

- **Statement 1 is correct:** Regional Rural Banks (RRBs) were established following the recommendations of the Working Group on Rural Banks headed by M. Narasimham in 1975. The committee recommended the creation of specialised banking institutions that could combine the local familiarity of cooperative institutions with the professional management and financial strength of commercial

banks. Consequently, RRBs were established under the Regional Rural Banks Act, 1976 to serve the credit needs of rural areas.

- **Statement 2 is correct:** RRBs have a unique ownership structure. Their share capital is jointly held by the Central Government (50%), the concerned State Government (15%), and the sponsoring commercial bank (35%). This arrangement ensures both public sector support and professional banking expertise in their functioning.
- **Statement 3 is incorrect:** RRBs are regulated by the Reserve Bank of India (RBI) under the Banking Regulation Act, 1949, and the Regional Rural Banks Act, 1976. NABARD acts as the principal institution for supervision, development, inspection, and refinancing support, but it is not the statutory regulator under the Banking Regulation Act.
- **Statement 4 is correct:** In line with their developmental mandate, RRBs are required to allocate 75% of their Adjusted Net Bank Credit (ANBC) to Priority Sector Lending (PSL). This target is significantly higher than that applicable to most commercial banks and reflects the objective of directing institutional credit towards agriculture, small enterprises, weaker sections, and other priority sectors in rural areas.

Q 126. Ans. B

Explanation:

- **Statement 1 is correct:** The southernmost point of the Indian Union is Indira Point (Great Nicobar Island), which extends up to approximately 6° 45' N latitude. The southernmost point of the Indian mainland is Kanyakumari, located at about 8° 4' N latitude.
- **Statement 2 is incorrect:** The Maldives is not separated from the Indian mainland by the Palk Strait. The Palk Strait separates the Indian mainland (Tamil Nadu coast) from Sri Lanka. The Maldives lies to the south of the Lakshadweep Islands and is separated from Minicoy, the southernmost island of Lakshadweep, by the Eight Degree Channel (8° N).

Q 127. Ans. C

Explanation:

- **Statement 1 is correct:** The Pir Panjal Range is the longest and the most important range of the Lesser Himalayas (also known as the Himachal range). It stretches from the Jhelum River to the upper Beas River for over 300 km.

- **Statement 2 is incorrect:** Geographically, the outer-most southern range of the Himalayas is the Shiwalik Range (Outer Himalayas). The Lesser Himalaya lies north of the Shiwalik range and south of the Great Himalayas (Himadri). Therefore, the Lesser Himalaya lies north, not south of the Shiwalik range.
- **Statement 3 is correct:** The Pir Panjal range forms the outer southern wall of the Kashmir Valley. By running parallel to the Great Himalayan range, it effectively separates the Kashmir Valley from the Outer Himalayas and the Jammu hills.

Q 128. Ans. A

Explanation:

- The Saltoro Ridge is a major physiographic feature of the Karakoram Range.
 - ✓ **Geographical Location:** It is located in the heart of the Karakoram Range, trending northwest-southeast just southwest of the massive Siachen Glacier.
 - ✓ **The Actual Ground Position Line (AGPL):** The ridge serves as a strategic natural wall that forms part of the Actual Ground Position Line (AGPL), dividing Indian-controlled territory in the Ladakh union territory from Pakistani-controlled regions (Gilgit-Baltistan).
 - ✓ **Strategic Significance:** During India's military deployment Operation Meghdoot in 1984, the Indian Armed Forces successfully established control over the key mountain passes along the Saltoro Ridge (including Sia La, Bilafond La, and Gyong La), giving India a dominant commanding position overlooking the valleys to the west.

Q 129. Ans. C

Explanation:

- The depression between the Rajmahal Hills (eastern edge of Chotanagpur Plateau) and the Meghalaya Plateau was formed due to a tectonic fault created during the north-eastward movement of the Indian Plate.
- After this rift/depression formed, it did not remain empty. Over geological time, it was gradually filled by massive alluvial sediments brought down by numerous rivers, most notably the mighty Ganga and Brahmaputra river systems. This process is known as fluvial deposition (river-based deposition).

Q 130. Ans. A

Explanation:

- Hydrological Unit Code (HUC) is the globally recognized standard scalar framework used to classify, define, and map watersheds based on their physical size and drainage area.
- **Originally developed by the United States Geological Survey (USGS) and widely adapted in global hydrological studies, the HUC system organizes watersheds into a nested, hierarchical numerical matrix. The system uses a sequence of numbers (digits) to represent varying scales of drainage basins:**
 - ✓ 2-digit HUC: Region (Largest)
 - ✓ 4-digit HUC: Sub-region
 - ✓ 6-digit HUC: Basin
 - ✓ 8-digit HUC: Sub-basin
 - ✓ 10-digit HUC: Watershed
 - ✓ 12-digit HUC: Sub-watershed (Smallest)
- **In India, a similar scalar hierarchy is maintained by the All India Soil and Land Use Survey (AISLUS), which categorizes drainage systems from largest to smallest as:** Water Resource Region → Basin → Catchment → Sub-catchment → Watershed.
- The runoff coefficient is a dimensionless numerical parameter used in hydrology (specifically the Rational Method) to calculate how much rainfall will turn into surface runoff based on surface permeability.

Q 131. Ans. C

Explanation:

- **Statement 1 is correct:** Iqbal strongly condemned a fatalistic approach to life and the passive acceptance of circumstances. He believed that human beings are co-creators with the divine and must actively shape their own destinies rather than submissively yielding to fate.
- **Statement 2 is incorrect:** Iqbal did not advocate an other-worldly or ascetic outlook. Instead, he rejected absolute asceticism and quietism (which he felt had led to the decline of Eastern societies). He emphasised fulfilling one's spiritual and moral potential within the material world through active engagement, social responsibility, and worldly striving.
- **Statement 3 is correct:** A core pillar of Iqbal's philosophy is dynamism. Influenced by both Islamic thought and contemporary Western philosophers (like Henri Bergson and Friedrich Nietzsche), he

viewed the universe as a continuous process of becoming. He emphasised that life is synonymous with constant change, struggle, creativity, and tireless activity.

Q 132. Ans. D

Explanation:

- **Statement 1 is incorrect:** Rana Sanga defeated Sultan Mahmud Khalji II (not Daulat Khan Lodi) of Malwa in the historic Battle of Gagron (1519). Daulat Khan Lodi was actually the governor of Lahore (Punjab) under the Delhi Sultanate who invited Babur to invade India to overthrow Ibrahim Lodi.
- **Statement 2 is correct:** Rana Sanga was the head of a formidable Rajput confederacy. Nearly all major Rajput chiefs—including Silhadi of Raisen, Medini Rai of Chanderi, Hasan Khan Mewati, Maldev Rathore of Marwar, and the rulers of Amber and Gwalior—owed allegiance to him and rallied under his banner before the Battle of Khanwa.
- **Statement 3 is correct:** Following his decisive victories over Ibrahim Lodi's forces in the Battles of Khatoli and Dholpur, Rana Sanga extended his military sway well into western Uttar Pradesh. His sphere of influence reached up to a small rivulet near Bayana (Pilya Khal), positioning his power within striking distance and directly adjacent to the regions surrounding Agra.

Q 133. Ans. A

Explanation:

- Babur declared the conflict against Rana Sanga as a jihad (holy war) to mobilise support against the Rajput confederacy.
- The battle in question is the Battle of Khanwa (1527), fought between Babur and Rana Sanga.
- Babur framed the war in religious terms to strengthen his position and legitimise his rule in India.
- This helped him consolidate Mughal power after his victory at Panipat (1526).

Q 134. Ans. B

Explanation:

- The rulers of the Vijayanagara Empire (1336–1646) were great patrons of literature, art, and architecture. While they maintained a multi-lingual court that patronised regional South Indian languages, primarily Telugu, Sanskrit, Kannada, and Tamil, the highest degree of royal favour, state documentation, and literary output was concentrated in Sanskrit and Telugu.

- ✓ **Sanskrit:** It remained the primary language for royal inscriptions, statecraft, philosophy, and religious texts. Prominent scholars like Sayana (who wrote commentaries on the Vedas) and his brother Madhava Vidyananya (the spiritual guide to the empire's founders) produced monumental works in Sanskrit under royal patronage.

- ✓ **Telugu:** The Vijayanagara period is universally considered the “Golden Age of Telugu Literature.” The language transitioned from being a regional dialect to the primary language of courtly poetry and administrative records.

Q 135. Ans. B

Explanation:

- The “Chain of Justice” (Zanjir-i-Adl) was installed by Jahangir at Agra Fort.
- It was a golden chain fitted with bells, running from the fort to the riverbank.
- Any aggrieved person could pull the chain to directly attract the emperor's attention and seek justice without intermediaries.
- This reflected Jahangir's emphasis on accessible and swift justice and his image as a ruler concerned with public grievances.
- Jahangir described it in the very first chapter of his autobiography, the Tuzuk-i-Jahangiri (Memoirs of Jahangir).
- According to Jahangir's memoirs, the chain was made of pure gold, was 30 gaz (yards) long, and was fitted with exactly 60 bells.
- It connected the Shah-burj of the Agra Fort to a post on the banks of the Yamuna River.

Q 136. Ans. B

Explanation:

- The Reserve Bank of India revised the Priority Sector Lending (PSL) norms for Small Finance Banks (SFBs) from FY 2025-26, reducing the mandatory PSL target from 75% of Adjusted Net Bank Credit (ANBC) or Credit Equivalent of Off-Balance Sheet Exposure (CEOBE), whichever is higher, to 60%. The objective of this change is to provide SFBs with greater operational flexibility in deploying their resources while preserving their developmental role.
- Small Finance Banks were established primarily to extend financial services to underserved segments such as small and marginal farmers, micro and small enterprises, and low-income households. However,

the very high PSL requirement often constrained their ability to diversify loan portfolios and manage risks efficiently. The revised norms allow SFBs to allocate a larger share of their funds to commercially viable non-priority sectors, particularly secured lending segments, thereby improving profitability, asset quality, and business sustainability.

Q 137. Ans. A

Explanation:

- **Statement 1 is correct:** DRR Dhan 100 (Kamala) and Pusa DST Rice 1 are among India's first genome-edited rice varieties developed using CRISPR-Cas-based genome-editing technology. These varieties were developed to improve traits such as yield, climate resilience, and stress tolerance without introducing genes from unrelated species. Their development marks a significant milestone in the application of advanced biotechnology for crop improvement in India.
- **Statement 2 is correct:** Unlike conventional genetically modified (GM) crops, these genome-edited rice varieties do not contain foreign or transgenic DNA. Instead, they involve precise modifications within the plant's own genome. Consequently, genome-edited plants developed through certain categories of genome editing are treated differently from transgenic GM crops under India's regulatory framework.
- **Statement 3 is incorrect:** In 2022, the Government of India exempted SDN-1 and SDN-2 genome-editing techniques from the stringent biosafety assessment process applicable to genetically modified organisms. However, this exemption is not applicable to all organisms. The relaxation was introduced primarily for certain genome-edited plant varieties, and the regulatory treatment of genome-edited animals, microorganisms, and other organisms continues to be governed by separate considerations and approvals.

Q 138. Ans. B

Explanation:

- **In Bihar, 30 out of the 38 districts have headquarters that share the same name as the district itself. The remaining 8 districts have headquarters with different names. These exceptions are Bhojpur (headquarters: Arrah), Saran (Chhapra), Vaishali (Hajipur), Kaimur (Bhabua), Rohtas (Sasaram), Nalanda (Bihar Sharif), East Champaran (Motihari), and West Champaran (Bettiah).**

Q 139. Ans. B

Explanation:

- **Pair 1 is correct:** Dr. Rajendra Prasad, popularly known as "Ajatshatru" for his ability to maintain friendly relations with all and as "Desh Ratna" (Jewel of the Nation) for his outstanding service to the country, was born on 3 December 1884 in Ziradei village of present-day Siwan district. Deeply influenced by Mahatma Gandhi during the Champaran Satyagraha, he abandoned his successful legal practice in 1920 to join the Non-Cooperation Movement. After Independence, he served as the first Minister of Food and Agriculture in the Interim Government of India from 1946 to 1947. He later became the first President of India and remains the longest-serving President, holding office from 1950 to 1962. In recognition of his immense contribution to the nation and his lifelong dedication to public service, he was awarded the Bharat Ratna in 1962.
- **Pair 2 is incorrect:** Vaidyanath Mishra, popularly known by his pen name Nagarjun (and Yatri in Maithili literature), was a renowned Hindi and Maithili poet, novelist, and social activist. He was born on 30 June 1911 in Satlakha, Bihar. A gifted scholar from an early age, he acquired deep knowledge of several classical languages, including Sanskrit, Pali, and Prakrit. In 1935, he traveled to Sri Lanka, where he embraced Buddhism and became a Buddhist monk, adopting the name Nagarjun.
- **Pair 3 is incorrect:** Anugrah Narayan Sinha, popularly known as "Bihar Vibhuti", was born on 18 June 1887 in Gaya. After completing his early education in Gaya, he obtained an M.A. in History from the University of Calcutta in 1914. He began his career as a history professor at T.N.J. College, Bhagalpur, where he actively participated in flood relief efforts for the local population. In 1916, he joined the legal profession by enrolling at the Patna Bar. Entering public life, he was elected to the Central Legislative Council in 1935 from the Sahabad-cum-Patna constituency. In 1937, he became a member of the Bihar Legislative Assembly and was appointed the Deputy Premier and Finance Minister of Bihar Province.
- **Pair 4 is correct:** Jayaprakash Narayan (JP), popularly known as "Lok Nayak" (People's Leader), was born on 11 October 1902 in Sitab Diyara. He left Patna College shortly before his examinations and enrolled at Bihar Vidyapeeth, founded by the

Indian National Congress. During the Quit India Movement, he displayed remarkable courage by escaping from Hazaribagh Central Jail on Diwali night after scaling a 17-foot wall while the guards were distracted by celebrations. Following his escape, he organized Azad Dasta (Freedom Brigade), a guerrilla group aimed at intensifying the struggle against British rule. In his 1959 book, *A Plea for Reconstruction of Indian Polity*, he proposed a four-tier system of governance based on village, district, state, and union councils, reflecting his vision of grassroots democracy and people-centered administration.

Q 140. Ans. B

Explanation:

- **Simple distillation** is the best method to separate a mixture of two miscible liquids that boil without decomposing and have a difference in their boiling points of greater than 25 Kelvin.
- **Filtration:** This method is used to separate insoluble solid particles from a liquid (for example, separating sand from water).
- **Separating funnel:** This apparatus is used to separate immiscible liquids, which are liquids that do not mix together (such as oil and water). They separate into distinct layers based on their varying densities.
- **Fractional distillation:** This process is specifically required when the difference in boiling points between two miscible liquids is less than 25 K. It utilizes a fractionating column to provide extra surface area for repeated condensation and vaporization, allowing for the precise separation of liquids with very close boiling points (for example, separating the different gases found in air or separating fractions of crude oil).

Q 141. Ans. D

Explanation:

- Metal-organic frameworks (MOFs) are highly porous crystalline materials composed of metal ions connected by organic molecules. You can think of them as microscopic sponges or molecular scaffolds.
- Because of their extraordinarily high internal surface area and tunable pore sizes, MOFs are exceptionally efficient at trapping, separating, and storing specific gas molecules. One of their most significant modern applications is in carbon capture technology, where they are used to scrub

carbon dioxide from industrial emissions to combat climate change. They are also heavily researched for safely storing alternative fuels like hydrogen and methane.

Q 142. Ans. C

Explanation:

- **Statement 1 is correct:** NISAR stands for NASA-ISRO Synthetic Aperture Radar. As the name suggests, it is an Earth-observation satellite that carries radar systems operating in two distinct microwave frequencies: the L-band and the S-band.
- **Statement 2 is correct:** The satellite is a joint venture. NASA provided the L-band radar, the high-rate communication subsystem, GPS receivers, and the large deployable reflector antenna. ISRO provided the S-band radar, the spacecraft bus, and the launch vehicle.
- **Statement 3 is incorrect:** NISAR is not placed in a geostationary orbit. It is designed to operate in a sun-synchronous Low Earth Orbit (LEO) at an altitude of approximately 747 kilometers. A low polar orbit is absolutely essential for an Earth-mapping satellite so it can sweep over the entire planet's surface every 12 days. Geostationary orbits are much higher (about 35,786 km) and remain fixed over one specific location, which would defeat the purpose of global mapping.

Q 143. Ans. C

Explanation:

- **Pair 1 is correct:** GlowCas9 is an engineered, bioluminescent version of the Cas9 enzyme developed by scientists at the Bose Institute in Kolkata. It emits light when the edited protein assembles correctly inside the cell, allowing researchers to visually track and monitor the gene-editing process in real time without harming the living cells.
- **Pair 2 is correct:** AlphaGenome is an advanced AI model developed by Google DeepMind. It is designed to analyze massive sequences of DNA to predict the functional and molecular effects of genetic mutations, specifically focusing on how variations in the noncoding regions of the human genome impact gene regulation and disease.
- **Pair 3 is correct:** BioEmu-1 (Biomolecular Emulator-1) is a deep learning AI model developed by Microsoft. Instead of just predicting a single static structure of a protein like earlier models, it generates thousands of structural predictions to map out the complete ensemble of different

dynamic conformations a protein can adopt, which is critical for understanding protein function and accelerating drug discovery.

Q 144. Ans. D

Explanation:

- **Statement 1 is correct:** Under the Special Assistance to States for Capital Investment (SASCI) scheme, there is a specific provision aimed at developing iconic tourist centres across the country to global standards to boost the overall tourism sector and local economies.
- **Statement 2 is correct:** Since this is a central assistance scheme providing 50-year interest-free loans to state governments, the actual implementation and execution of the approved tourism projects on the ground are carried out by the respective State Governments.
- **Statement 3 is incorrect:** The SASCI scheme itself is administered by the Ministry of Finance (specifically the Department of Expenditure). The specific proposals for the iconic tourist centres are evaluated, approved, and promoted by the Ministry of Tourism, not the Ministry of Culture.

Q 145. Ans. D

Explanation:

- **Statement 1 is correct:** The Economic Survey 2025-26 projects that India's real GDP growth will remain robust, forecasting a growth rate in the range of 6.8 to 7.2 percent for the financial year 2026-27 (FY27).
- **Statement 2 is correct:** The industrial sector showed a strong structural recovery. The Survey specifically highlights that manufacturing emerged as a key growth engine, with gross value added (GVA) growth accelerating to 7.72 percent in Q1 and reaching 9.13 percent in Q2 of FY26.
- **Statement 3 is correct:** Prudent fiscal management was a key theme of the Survey. It notes that the government set a target to further decline the central government's fiscal deficit to 4.4 percent of GDP for FY26, fulfilling a long-term goal of fiscal consolidation.
- The Economic Survey 2025-26 also emphasized the structural transformation in manufacturing, noting that medium and high-technology activities now account for 46.3 percent of India's total manufacturing value added. It also highlighted that the Production Linked Incentive (PLI) schemes generated massive employment and investments across 14 strategic sectors.

Q 146. Ans. A

Explanation:

- **Statement 1 is correct:** An isoquant curve (from "iso" meaning equal and "quant" meaning quantity) is a contour line drawn through the set of all possible combinations of two inputs, typically Labor and Capital, that produce the exact same level of total output. It is the producer's equivalent of a consumer's indifference curve.
- **Statement 2 is incorrect:** Isoquant curves are universally downward sloping from left to right (negatively sloped), not upward sloping. This structural property exists because the inputs are assumed to be substitutable. If a firm decides to use more labor, it must systematically use less capital to keep the overall output level identical. If the curve were upward sloping, it would imply that increasing both labor and capital produces the exact same output, which logically violates the basic economic assumption that adding more resources increases total production.
- **Other foundational properties of isoquant curves:**
 - ✓ They are convex to the origin due to the diminishing Marginal Rate of Technical Substitution.
 - ✓ Two isoquant curves can never intersect each other.
 - ✓ A higher isoquant curve positioned further to the right always represents a higher level of aggregate output.

Q 147. Ans. D

Explanation:

- During the colonial era, the British government never made any sincere or systematic attempt to estimate India's national and per capita income to understand the economic condition of the country. These early estimates were carried out entirely by individual economists, leaders, and researchers.
- **Dadabhai Naoroji:** He was the pioneer in this field. He prepared the very first estimate of national income for the year 1867-68 in his famous book *Poverty and Un-British Rule in India*.
- **William Digby:** He published his estimates for the year 1899.
- **R.C. Desai:** He estimated consumer expenditure and national income for the period between 1931-32 and 1940-41.
- **Findlay Shirras:** He calculated national income estimates for several years, including 1911, 1922, and 1931.

- Dr. V.K.R.V. Rao is credited with making the first truly scientific and systematic estimation of India's national income for the year 1931-32. The methods established by V.K.R.V. Rao laid the foundation for modern national income accounting in independent India.

Q 148. Ans. D

Explanation:

- **Statement 1 is incorrect:** Correcting the weaknesses that had developed in the balance of payments is a short-term objective classified under stabilisation measures, not structural reforms. Its aim was to restore the immediate macroeconomic position by maintaining sufficient foreign exchange reserves.
- **Statement 2 is correct:** Enhancing long-term economic efficiency and international competitiveness is the central aim of structural reform policies, which are long-term measures implemented through liberalisation, privatisation, and globalisation.
- **Statement 3 is correct:** Removing the rigidities in various segments of the domestic economy is the defining method of structural reform policies, undertaken to make the economy more efficient and globally competitive.
- **Statement 4 is incorrect:** Keeping rising domestic prices and inflation under control is a short-term goal classified under stabilisation measures, alongside correcting the balance of payments, and is therefore not a structural reform policy.

Q 149. Ans. B

Explanation:

- **India's Balance of Payments has two broad parts:** the Current Account, which records trade in goods and services, income, and current transfers; and the Capital (and Financial) Account, which records cross-border movements of investment, loans, and other capital. Each transaction is classified below.
- **Statement 1 is recorded in the Current Account:** Software services exports by an Indian IT firm are an export of services. Trade in services is part of the invisibles in the current account, so this is a current account credit.
- **Statement 2 is recorded in the Capital Account, not the Current Account:** The purchase of shares of a British company by an Indian resident is an outward portfolio investment, that is, the acquisition of a foreign financial asset. Investment flows of this kind belong to the capital and financial account.

- **Statement 3 is recorded in the Current Account:** Remittances sent home by Indian workers abroad are private (unilateral) current transfers, with no corresponding flow of goods, services, or assets in return. Such transfers are recorded under the current account.
- **Statement 4 is recorded in the Capital Account, not the Current Account:** The repayment of the principal of an External Commercial Borrowing reduces India's external liabilities, which is a capital and financial account transaction. Note the distinction: the interest paid on the ECB would be a current account item (investment income), but the repayment of the borrowing itself is a capital account flow.

Q 150. Ans. B

Explanation:

- **Statement 1 is incorrect:** The issuance of Commercial Paper is not open to "any company regardless of its net worth." Every CP issue must carry a minimum credit rating from a recognised rating agency, and the issuer's fund-based facilities from banks, AIFIs, or NBFCs must be classified as Standard at the time of issue, so eligibility is conditional, not universal. The older framework explicitly required a minimum tangible net worth of about ₹4 crore for a corporate issuer, while the RBI's 2024 Master Direction keeps a stated net-worth floor of ₹100 crore only for "any other body corporate," co-operative societies, and LLPs, not for companies as such.
- **Statement 2 is correct:** Certificates of Deposit issued by scheduled commercial banks (excluding RRBs and Local Area Banks) have a maturity of 7 days to a maximum of one year, whereas those issued by eligible All-India Financial Institutions can be issued for a minimum of one year and up to three years.
- **Statement 3 is incorrect:** The Call and Notice Money market is restricted to banks and Primary Dealers, who alone may participate as both borrowers and lenders. Non-bank institutions other than Primary Dealers, including NBFCs and corporates, are not permitted to participate. So NBFCs are not regular participants in the call money market.
- **Statement 4 is correct:** Treasury Bills are issued by the Reserve Bank of India on behalf of the Central Government (in tenors of 91, 182, and 364 days) and are zero-coupon instruments, that is, they carry no interest, being issued at a discount to face value and redeemed at par.

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Tower, Boring Road
Chauraha, Patna,
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Naya Tola

KP Yadav
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